Lecture Transcript by Slide

Week 11 · Chapter 8 practice and Chapter 7: Heart Blocks · 1 h 46 min
Auto-transcribed, so expect occasional errors, especially in technical terms. Timestamps are positions in the recording.

0:00:00 Ventricular rhythms practice (screen share not recorded)

0:56:16 Chapter 7: Heart Blocks (title slide)

0:56:35 AV Heart Blocks

1:06:20 Mechanism of First Degree Heart Block

1:06:51 Rules for First Degree Heart Block

1:07:53 First Degree Heart Block (ECG strip)

1:12:00 Mechanism of Wenckebach (Type I Second-Degree Heart Block)

1:14:33 Rules for Wenckebach

1:16:22 Wenckebach (ECG strip)

1:23:22 Mechanism of Type II Second Degree Heart Block

1:24:58 Blocked P Waves

1:29:10 Type II Second-Degree Heart Block (ECG strip)

1:31:04 Conduction Ratios in Type II Second Degree Heart Block (2:1, 3:1)

1:31:57 Conduction Ratios in Type II Second Degree Heart Block (4:1, variable)

1:33:15 Type II Second Degree Heart Block with a Prolonged PRI

1:33:37 Rules for Type II Second Degree Heart Block

1:34:23 Rules for Type II Second Degree Heart Block (continued)

1:35:02 Mechanism of Wenckebach (revisited)

1:35:30 Mechanism of Complete Heart Block (CHB)

1:36:52 A–V Dissociation in CHB (a, b)

1:38:15 A–V Dissociation in CHB (c)

1:44:23 Rules for Complete Heart Block

1:45:04 Chapter 7 Practice Strips

1 Ventricular rhythms practice (screen share not recorded)

0:00:00 – 0:56:16

0:00:04 Hmm! Michael V Hyland: Seeing her lord Michael V Hyland: one more than the little you know. Michael V Hyland: So I'm not ordering Michael V Hyland: usual amount of order. The book does heart blocks Michael V Hyland: before they do ventricular rhythms. I much prefer to do ventricular rhythms than heart blots. I told you that eventually we were gonna get to some difficult content. Michael V Hyland: That difficult content will be today. Alright. So I'm going to start with ventricular rhythms. So these our rhythms that come from Michael V Hyland: ventricular muscle cells.

0:01:20 Michael V Hyland: or it could also come from somewhere along the pathway, but down in the ventricles. like the the bundle branches and the and brackenji fibers. Michael V Hyland: So anywhere in the ventricle can produce a single beat

0:01:38 Michael V Hyland: or an entire rhythm. Michael V Hyland: These beats. Michael V Hyland: for the most part, are going to be caused by irritability.

0:01:54 Michael V Hyland: So we're gonna have irritability Michael V Hyland: which Michael V Hyland: run Michael V Hyland: and cause a single beat. Michael V Hyland: And in this case, so remember, we've talked about Pacs. We've talked about Pjcs. Now we're going to talk about PVC's premature ventricular complexes.

0:02:24 Michael V Hyland: It can also, this irritability Michael V Hyland: cause an entire rhythm. Remember fastest rate, wind. So if one irritable spot fires repetitively, and it fires at a rate faster than the inherent pacemaker it will take over Michael V Hyland: as the pacemaker.

0:02:48 Michael V Hyland: The other thing that we're going to talk about here is there is one rhythm that is not irritable. Michael V Hyland: it's a backup pacing rhythm. And so if the sa node fails Michael V Hyland: and the AV note fails, the last backup pacemaker here is gonna be the ventricular escape rhythm or Ver. So just like we had junction escape, which is the primary backup. We're gonna talk today about ventricular escape. Michael V Hyland: which is the last backup pacemaker.

0:03:25 Michael V Hyland: And then, before I start talking about these rhythms, the other thing that I want to talk about is because these beats are coming from the ventricles themselves.

0:03:38 Michael V Hyland: When we talk about super ventricular beats, beats, beats that come from above the ventricle. Those Michael V Hyland: make the ventricles depolarize through the normal, the flow of electricity goes over the normal nerve pathway Michael V Hyland: when we have a muscle cell in the ventricle. Now that stimulates a beat like where the green star is that impulse has to spread cell to cell. Michael V Hyland: It doesn't travel over a nerve.

Remember, when it travels over a nerve, it travels fast. Depolarization happens quickly. When we record a super ventricular beat it will have a narrow Qrs. Michael V Hyland: Now, because it has to spread cell to cell. It doesn't travel as fast. Depolarization takes longer when we record that depolarization on paper. Now, we're going to have a wide Qrs.

0:04:34 Michael V Hyland: so it's going to be wider. It's going to be point 1 2 s or greater. Michael V Hyland: 3 small boxes, or greater, because depolarization takes longer when it travels from cell to cell. The impulse travels from cell to cell Michael V Hyland: as opposed to traveling on a nerve.

0:04:56 Michael V Hyland: Alright. Michael V Hyland: So the first rhythm we're going to talk about here. And it's actually not a rhythm. It's a single beat. This is called premature ventricular, complex, or PVC. Michael V Hyland: So

0:05:11 Michael V Hyland: it's a single irritable focal point within the ventricle that fires Michael V Hyland: prematurely. We know that early beats are irritable beats Michael V Hyland: so premature that that concept stays the same and initiates a single ectopic complex. Michael V Hyland: This is a single beat, not a rhythm. So again, we still have to identify the underlying rhythm just like we did with pacs, just like we did with Bjc's.

0:05:43 Michael V Hyland: okay. Michael V Hyland: So the other thing that we're going to see here, besides a wide Qrs. Is this beat initiates in the ventricles, so there will be no P wave Michael V Hyland: in front of it, or after it. There will be no associated P. Wave with a premature ventricular complex.

0:06:11 Michael V Hyland: It will be a wide QRS. Without a p wave. and these QRS. Look wide. and usually are described as wide and bizarre.

0:06:24 Michael V Hyland: You'll see what I mean as we go through these. So I'm going to give you some examples of some PVC's. Michael V Hyland: So in the Michael V Hyland: a box. Michael V Hyland: we're looking at 2 beats from the underlying rhythm. followed by

0:06:45 Michael V Hyland: a QRS. Complex. So this QRS. Complex Michael V Hyland: early, because you can see that it's pretty close to the preceding beats. T wave. So it's early. There's no P wave in front of it. Michael V Hyland: It just drops down to a negative Michael V Hyland: r wave. There is no Q wave a negative r wave back to the S wave, and then the T wave goes after that. and then we have what we call that compensatory pause, and then the underlying rhythm kicks back in again. Michael V Hyland: So what's difficult about this is 2 things. Michael V Hyland: Number one. How do I measure it? How do I measure the width of this? Number 2? Where does the S wave end and the t-wave begin.

0:07:39 Michael V Hyland: So it's not exactly Michael V Hyland: an exact science. Michael V Hyland: So what we're doing here is we're looking for a little wiggle at the end of this. So the S wave actually stops right there. And this is the beginning of the T wave right there. So we're going to measure from that initial dropdown right to there.

0:08:06 Michael V Hyland: So that's going to be 1, 2, 3, maybe 3 and a half small boxes. Michael V Hyland: Everybody see what I did there. so I'll do the cursor again. So we said.

0:08:22 Michael V Hyland: We said that this is where the S wave ends and the T. Wave begins. So it's right on that line right there. Michael V Hyland: right on that line, and then the the negative deflection begins either halfway in between or right there. Michael V Hyland: So I'm going to count backwards from that line right there, 1, 2, 3, and I think it's 3 and a half, because it drops just before that line. You follow me. Now.

0:08:57 Michael V Hyland: remember, I'm still going to give you that one box grace period to get it right? Right? So you clearly see, there's no P wave there. The other thing that is generally identifiable about these PVC S is the R wave and the T wave Michael V Hyland: usually Michael V Hyland: notice, I haven't said there's a hundred percent rules yet. This semester. So I'm gonna say, usually 98, 99% of the time our wave team wave go in opposite directions. Michael V Hyland: So we're gonna not have a P wave. We're gonna have a wide, bizarre Qrs and R and T. Usually always go in the opposite direction. So R. Wave is down. T. Wave is up Michael V Hyland: in the next example. In the rhythm marked B.

0:09:47 Michael V Hyland: Again, we have 2 rhythm, 2 beats from the underlying rhythm, the T-wave. We have a short period before we get to this wide, bizarre QRS.

0:09:59 Michael V Hyland: You can see that there's no P. Wave. R. Wave is up, T. Wave is down, and then we have a compensatory pause. So this is an early beat. It's a wide beat again. You're looking for the little bend down here which occurs right there

0:10:17 Michael V Hyland: right there. Michael V Hyland: So this is. If you count one. Michael V Hyland: 2, 3. This is 3 small boxes. So this, the width of this Qrs is point 1 2 s.

0:10:33 Michael V Hyland: Okay. Michael V Hyland: it's going to take some adjustment on your guys behalf to get used to looking at these weird complexes and figuring out how to measure the width of the Qrs.

0:10:49 Michael V Hyland: Alright. Some more examples Michael V Hyland: again, usually always R and T in opposite directions. So the first one R is down T is up. Michael V Hyland: We would count this initial hump, so we got a little positive deflection followed by that. So this is a pretty wide Qrs

0:11:13 Michael V Hyland: in B we have a positively deflected R wave with a negatively deflected T wave. Michael V Hyland: and again we would measure from here to here.

0:11:26 Michael V Hyland: You see. Michael V Hyland: this has a little bit of a pause after it. A has a no pause after it goes right back into the regular rhythm. I'm not gonna really expect you to to tell me. Michael V Hyland: does it have a pause? Does it not have a pause?

0:11:43 Michael V Hyland: And I'm also not gonna expect you to tell me the name of it right? Michael V Hyland: A PVC. Without a pause, is called an interpolated PVC. Not going to test you on it. Don't expect you to Michael V Hyland: label that on the homework or on the test. Okay? Michael V Hyland: But you can see 4 different examples of how PVC's might appear in A, BC and D

0:12:15 Michael V Hyland: again, pretty straightforward. These are single beats caused by irritability. They're early beats. Why, we call them premature. Michael V Hyland: So 4 more different examples of what a PVC might look like

0:12:38 Michael V Hyland: in example. H. There's really no t-wave. We just have a wide and bizarre QRS. There's no p-wave with a wide and bizarre Qrs.

0:12:50 Michael V Hyland: And the other thing is is, remember, like I said, with Pacs or Pjcs. Michael V Hyland: I can give you a 6 s strip with one PVC. In it. I could give you a PVC. Or a 6 s strip with 2 PVC. S. In it. Michael V Hyland: or even 3 or 4 PVC. S. In it. Michael V Hyland: The difficult thing about ventricular rhythms is the PVC's can appear a lot of different ways and a lot of different patterns. So we're gonna cover all of these different patterns. Michael V Hyland: So for H, where would you measure to right underneath the isometric?

0:13:30 Michael V Hyland: Yeah, you're just gonna measure the total width of that complex. Michael V Hyland: So if you follow my cursor, it starts right on that line right there.

0:13:45 Michael V Hyland: and it ends right there. Michael V Hyland: 1, 2, 3, 4, Michael V Hyland: probably 4 and a half. Michael V Hyland: maybe even 5. So point 1 8.2 0 s. everybody comfortable with that

0:14:05 Michael V Hyland: in in example E, this is where it starts right there it starts right there, but it ends right here. You see the bend coming off. This is Tway, so you would measure from here Michael V Hyland: to the bend. Michael V Hyland: So you're always looking for that bend at the end of the S wave which tells you the start of the T wave. That's kind of an important trick in measuring the width of the QRS. Because it can be difficult.

0:14:40 Michael V Hyland: What about G. What about it? Michael V Hyland: So there I would start here. Michael V Hyland: and because I don't really see a full bend. Michael V Hyland: What I would do is just go straight across to the isoelectric line, and I would measure from the year to

the beginning Michael V Hyland: right there.

0:15:05 Michael V Hyland: Aye. these can be tricky. Michael V Hyland: So you gotta you gotta work at this a little bit. Okay? Michael V Hyland: So when a PVC occurs. Michael V Hyland: we usually get that compensatory pause. And again, I don't expect you to understand this in terms of Michael V Hyland: homework or testing.

0:15:28 Michael V Hyland: and that pause is usually 2 times the r to our interval. Okay. Michael V Hyland: PVC. That occurs in between a rhythm without a pause. is the interpolated PVC. That I was talking about. and that doesn't really break the regularity of the ribbon.

0:15:52 Michael V Hyland: So remember how we would do this one. With the compensatory pause we would say we would count the spacing between our waves Michael V Hyland: here, and this R. Wave is down, so we would count from there to there, and then there, to there. and then to there. So this rhythm would be regular with one ectopic B.

0:16:16 Michael V Hyland: This rhythm would also be regular with one at topic B. Remember how we classify rhythms in terms of regularity. Michael V Hyland: Okay. Michael V Hyland: so I told you I could give you a rhythm, a 6 s strip with more than one PVC. In it. And here's an example of that.

0:16:38 Michael V Hyland: And here is where you're going to need to know the terminology. Michael V Hyland: So Michael V Hyland: the PVC S appear on the third beat and the seventh beat. Michael V Hyland: Okay. But if we map this out Michael V Hyland: from here to here, and then here, to here, and here to here, those spacings are going to be the same. Remember, all you need is 2 different spaces Michael V Hyland: of R to R in a strip that are the same. Then we're labeling that rhythm as regular. Michael V Hyland: Don't forget that part right. I can find 2 r to our intervals in a 6 s strip that have the same space Michael V Hyland: that underlying rhythm is regular. So this rhythm is regular, interrupted by 2 tech topics.

0:17:38 Michael V Hyland: and when I look at those at topics they look exactly the same. Michael V Hyland: The width of the width of the Qrs is going to be the same, and so when they look the same, they're coming from the same focal point. So the same irritable, ventricular muscle cell generated both of those beats.

0:18:05 Michael V Hyland: So when they look the same, we're going to call them unifocal. PVC's unifocal Michael V Hyland: from the same focal point.

0:18:16 Michael V Hyland: So, looking at the spacing here, I'm pretty sure the interpretation of this rhythm is, gonna be normal sinus rhythm. If we mapped it all out, it's gonna be normal sinus rhythm with 2 PVC's that look the same, so the full interpretation would be normal sinus rhythm with 2 unifocal PVC's

0:18:42 Michael V Hyland: the next rhythm. Michael V Hyland: It's the same normal sinus rhythm. And it has 2 Pbc's in it. So again, this rhythm would be regular, because the spacing is going to be normal.

0:18:57 Michael V Hyland: regular with 2 Ec topics. Michael V Hyland: But these PVC's don't look the same one is positively deflected. One is negatively deflected. So that means they're coming from different Michael V Hyland: ventricular muscle cells. So more than one focal point. Michael V Hyland: So now we're going to interpret this rhythm as normal sinus rhythm with 2 multifocal meaning, more than one focal point. PVC's.

0:19:28 Michael V Hyland: You've got to get this terminology down, because if you tell me normal sinus rhythm with 2 PVC's. Michael V Hyland: I'm gonna take a point off.

0:19:39 Michael V Hyland: I'm expecting you to tell me these are multifocal. They don't look the same, or they're unifocal, they do look the same. So if you don't include that terminology, I'm going to, I'm going to deduct points.

0:19:59 Michael V Hyland: The other thing that I would expect you to add to this. Michael V Hyland: And again, because my grading is fair, I feel like my grading is more than fair throughout the semester that I'm expecting you now to measure the Qrs width of the underlying rhythm and the Qrs width of the PVC. So on the Qrs line. Michael V Hyland: you're going to start with the underlying Qrs. And then just write a slash Michael V Hyland: and then write the width of the PVC.

0:20:38 Michael V Hyland: And if you don't give me both, guess what I'm gonna do. Michael V Hyland: I'm going to take a point off. Michael V Hyland: I'm not adding an extra point because you gave it to me. I'm telling you what my expectation is so. If you don't give me both, I'm taking a point off

0:20:54 Michael V Hyland: trying to be very upfront and clear as to what my expectations are. Michael V Hyland: and if these PVC's the width of these 2 multifocal PVC's aren't the same. Michael V Hyland: then you would give me a third Q. Rs measurement. If they're the same, then you just give me 2

0:21:17 Michael V Hyland: all right again. This is not something I'm gonna test on either. But I think that this is this is this is important to know if you're in a job where you're gonna be monitoring. Ekgs, this is important to know. Michael V Hyland: So if you look at the first PVC, the R. Wave of that first PVC. Michael V Hyland: Right here. This R wave is connected to the T wave. We call that R on T phenomena.

0:21:48 Michael V Hyland: and that our aunty phenomenon can increase irritability. Michael V Hyland: and when we have r on T, it can cause the patient to go into a life threatening arrhythmia, either vfib ventricular fibrillation or ventricular attack of cardio. We're gonna cover both of those rhythms shortly today Michael V Hyland: where the other PVC. B. There's a short pause at the isoelectric line. There's no R. On T phenomenon on that one. Michael V Hyland: The more often this occurs.

0:22:24 Michael V Hyland: or sometimes it can occur right away that phenomenon is potentially life-threatening. Michael V Hyland: So if the underlying heart rhythm Michael V Hyland: is adequate like not you know what if it's adequate, like. Michael V Hyland: like the underlying rhythm, the heart rate is their 50 s. Or 60 s. Or 70 S. We would give drugs, ventricular arrhythmia drugs to suppress these PVC's Michael V Hyland: to try to prevent that phenomenon from happening. Okay. Michael V Hyland: 10. Not going to be on the homework, not going to be on a test. Just a potential life-threatening piece. Michael V Hyland: Alright. So what do you see on this rhythm strip?

0:23:14 Michael V Hyland: There's 2 PVC's in a row. Okay. so we call this a coupler. right? Like a couple holding hands. Oh, isn't that sweet? Michael V Hyland: Right? So we call this a couplet. So again, looking at that spacing. This rhythm is regular with 2 Ec. Topics.

0:23:39 Michael V Hyland: and then my interpretation of this rhythm would be normal sinus rhythm Michael V Hyland: with one couplet or one couplet of PVC's.

0:23:56 Michael V Hyland: So when we had 2 topics in a row, we call it a couplet Michael V Hyland: again. Got to know this terminology right unifocal multifocal couplet, we're not done Michael V Hyland: so. Now we have a rhythm with 4 PVC's in a row.

0:24:17 Michael V Hyland: The underlying rhythm is regular. Michael V Hyland: I would also venture to say that the spacing between the 4 PVC's is probably also regular. Michael V Hyland: but for the purposes of the homework and the test you would say, this rhythm is regular with 4 ectopic beats.

0:24:44 Michael V Hyland: Now. Michael V Hyland: when we have 3 or more PVC. S. In a row, we call it a short burst. Michael V Hyland: or a short run of ventricular tachycardia. Michael V Hyland: so these are ventricular beats. You can see that spacing is kind of narrow. If we actually Michael V Hyland: calculated the heart rate of the 4 PVC's, it would be above a hundred. Okay. Michael V Hyland: so my interpretation of this rhythm would be normal sinus rhythm Michael V Hyland: with a 4 beat run of Vtac Michael V Hyland: or a 4, Michael V Hyland: the first of V tech. So when we have 3 or more, we call it a run or a burst of ventricular tachycardia, v-tax fans for ventricular tachycardia.

0:25:37 Michael V Hyland: and when we get to ventricular tachycardia. This is going to what it looks like. Michael V Hyland: 4 beats in a row. It's going to be very similar to that. Michael V Hyland: That's a good question. I'm glad that you asked that question. So you're going to provide me the heart rate of the underlying rhythm. Michael V Hyland: So what do we have for spacing on this rhythm 510-14-1516, and 2 halves make 17. Michael V Hyland: So 1,500, divided by 17

0:26:21 Michael V Hyland: 88. So the heart rate's gonna be 88. Michael V Hyland: So the rhythm is regular. Michael V Hyland: with 4 ectopic beats, the heart rate is 88. You would still mark P. Waves that are upright and uniform for the underlying rhythm.

0:26:38 Michael V Hyland: we would still measure the Pr. I. For the underlying rhythm. Michael V Hyland: We would measure the width of the Qrs for the underlying rhythm and the width of the PVC's. They're all gonna be the same. So you're only gonna give me 2 for this rhythm, and then your interpretation is gonna be normal sinus rhythm with a 4 beat runa beat tack

0:27:00 Michael V Hyland: pretty clear. Be clear now while we're talking, but if you don't go back and practice this. it's gonna be a problem. I've been teaching this class for 25 years. Michael V Hyland: So if you don't go back and review within the next few days, and then another couple of days after that it's going to be a problem. Michael V Hyland: You got to put some work into these rhythms because there's a lot of terminology here, and a lot of different rhythms. All right, we're going to continue with our pattern of PVC's. What do we see in this rhythm?

0:27:37 Michael V Hyland: Well, the first beat Michael V Hyland: I see Michael V Hyland: is probably a beat for the underlying rhythm. It's a narrow QRS. Michael V Hyland: I only see half a T. Wave. and then I go right into a PVC. R. Wave down here. T. Wave here, see this bend way up here at the cap. That's where the S. Wave would be. Michael V Hyland: So you're not going to measure the T-wave. You're going to go from the peak here, right to there in terms of measuring that. Qrs. Michael V Hyland: okay? Michael V Hyland: And what I'm seeing is an underlying beat, followed by a PVC. An underlying beat, followed by a PVC. So ultimately, what I said is, what's this? What do you see here? I see a PVC. Every other beat Michael V Hyland: underlying rhythm. PVC. Underlying rhythm. PVC. Every other beat. So when you see PVC's every other beat, we're going to label it as Bigeemony

0:28:35 Michael V Hyland: now we still have to to figure out the underlying rhythm. You can't just put bigemony. Michael V Hyland: So if you look at this.

0:28:46 Michael V Hyland: and we also don't count PVC. S. In the heart rate. The reason why we don't count PVC. S. In the heart rate is, we don't know if they refuse. Sometimes a premature ventricular complex will actually eject blood from the ventricle. Other times it doesn't eject

blood from the ventricle, so we can't count on them to contribute to cardiac output. So we do not count them in the heart rate.

0:29:15 Michael V Hyland: So what we would do here Michael V Hyland: is I'm counting the number of small boxes between the underlying R waves. So let's do that. 5, 1015, 20, Michael V Hyland: 2324, 25, and 2 halves make 26, and I'm pretty sure it's going to be 26 all the way across. So I would say that this rhythm is regular Michael V Hyland: with an ectopic beat. Every other.

0:29:51 Michael V Hyland: My heart rate is going to be 1,500, divided by 26. Michael V Hyland: That's going to be a low heart rate.

0:30:06 Michael V Hyland: Who's got the heart rate? 58. Michael V Hyland: So then I'm going to go to my P. Waves. There are P. Waves that are operating uniform. We got to measure the Pr. I. We got to measure the width of the Q. Rs. Both for the underlying rhythm and for the PVC.

0:30:31 Michael V Hyland: And then my interpretation stay connected to your data points. This rhythm is regular with a heart rate of 58. What's the underlying rhythm?

0:30:45 Michael V Hyland: Sinus bradycardia Michael V Hyland: with bigeemony? That's how you would interpret this rhythm, Sinus Bradycardia with Bigeemony or Bigeemony of PVC's Michael V Hyland: so in this case do. The PVC's all have to be uniform.

0:31:06 Michael V Hyland: They generally are. Yeah, they generally are Michael V Hyland: questions on this. Michael V Hyland: Okay, well, if I can have a PVC every other beat.

0:31:21 Michael V Hyland: I can have a PVC. Every third B, so we would label this as triggering. Michael V Hyland: Now, remember. Michael V Hyland: we said, if you can find 2 R to R's throughout the strip. The underlying rhythm is regular. so do you think so? The first beat is there, followed by a PVC. The third beat and the fourth beat. Michael V Hyland: Count the number of small boxes there right. and then

0:31:58 Michael V Hyland: the sixth and the seventh beat. Count the number of small boxes there. I'll bet you that are the same. So this rhythm is regular with an ectopic every third beat.

0:32:16 Michael V Hyland: What's my spacing here? 10, 1112, and 2 halves make 13. So I got 1,500 divided by 13. Michael V Hyland: It's gonna be 107, 1 15.

0:32:39 Michael V Hyland: So I'm still looking at my P. Waves, which are upright in uniform. Michael V Hyland: I'm measuring my Pri, I'm measuring the width of the underlying rhythms. QRS. And the PVC's. Michael V Hyland: These PVC's will all measure out the same. And then

again, I need to stay connected. The rhythm's regular with ectopics, but the heart rates 1 15, Michael V Hyland: and I got a PVC. Every third beat. What's my interpretation. sinus Michael V Hyland: Pachaccardia with trigemine.

0:33:15 Michael V Hyland: So when we see the regularity, can we say regular with you? Could you email, the other thing you could do is because there's foreign topics here. You could just say regular, with foreign topics, regular with multiple, like topics. Michael V Hyland: Right?

0:33:33 Michael V Hyland: We're pretty flexible with that. I really just, you know. Michael V Hyland: and I haven't even taken off if you just say regular. Michael V Hyland: And you don't say anything about egg topics. I haven't been marking it wrong, but technically, I probably should be. So I'd like to see you guys start, including regular Webc topics.

0:33:54 Michael V Hyland: Well. one more. Why not? Michael V Hyland: So we can have a PVC. Every fourth week, and we would call that quadrgeem.

0:34:07 Michael V Hyland: Do I need to go through the details here? Or you guys think you got it Michael V Hyland: right Michael V Hyland: again? The underlying rhythms gonna be regular cause. The spacing is gonna the art. Our spacing for the underlying rhythms gonna be regular. So regular with that topics

0:34:25 Michael V Hyland: number of small boxes under the regular part in the 1,500 P. Waves, measuring pr, i. Measuring QRS. Both Q. Rss. Right?

0:34:38 Michael V Hyland: And this still looks like it's gonna be sinus tag Michael V Hyland: with quadrgemony. Michael V Hyland: Okay? Alright Michael V Hyland: moving on. So rules for premature ventricular complex. Michael V Hyland: the regularity cause. Remember, these are single beats at topic beats. So the regularity for this is really gonna depend on the underlying rhythm. These topics disrupt it. Michael V Hyland: but it still can be regular Michael V Hyland: not to mention we can have PVC's in other rhythms. Michael V Hyland: You could be in a fib and have a PVC. You could be in a junctional rhythm and have a PVC. Right?

0:35:20 Michael V Hyland: So not that I'm gonna do that to you. But you need to be aware of that. The rate is going to depend on the underlying rhythm. We do not count PVC's in the rate. Michael V Hyland: there will not be A. P. Wave for the PVC. But there may be there should be P. Waves in the underlying rhythm.

0:35:42 Michael V Hyland: Every once in a while we may see A. P. Wave front of a PVC. But it's not related to that. PVC. I'm going to talk about that when we talk about blocks today.

0:35:55 Michael V Hyland: So I'll revisit that in a little bit. Michael V Hyland: Pri, we're still going to measure the Pri for the underlying rhythm. Michael V Hyland: The QRS. We're going to measure the underlying QRS. And Michael V Hyland: the QRS. For the PVC. Michael V

Hyland: And, as I already stated, T. Wave usually goes in the opposite direction of the R way. So these PVC's don't have a P. Wave. They're wide and bizarre. R and T. Go in opposite directions.

0:36:27 Michael V Hyland: And then here's just another example of any guesses on the interpretation of this rhythm. Look at your spacing. Does that look like a fast spacing or a regular spacing. Michael V Hyland: Yeah, it's about 14, maybe 14 and a half, so maybe right at 103.

0:36:51 Michael V Hyland: You don't trust me. Michael V Hyland: 7107, Michael V Hyland: right. Do those? PVC's look the same? Michael V Hyland: So my interpretation is going to be Michael V Hyland: sinus tachacardia with Michael V Hyland: 2 unifocal PVC's good. Okay.

0:37:15 Michael V Hyland: all right? Now, we're Gonna talk about ventricular tachycardia. Michael V Hyland: So again, this is a single irritable focal point within the ventricles, but now it fires rapidly, repetitively, at a rapid rate, and its rapid rate is faster than the inherent pacemaker. So it takes over

0:37:36 Michael V Hyland: the rate range for this. V tack is 1 50 to 2 50. So it's fast. So what are we talking about? One PVC. Followed by another, followed by another, followed by another, followed by another. Michael V Hyland: so there will not be any P. Waves. It'll be wide, bizarre. R and T. Usually go in opposite directions. So let's look at the rules. Then we'll look at the rip Michael V Hyland: so it's usually regular. There are times where it can be slightly irregular. meaning that R to R is a smidge longer Michael V Hyland: than a box and a half. Remember what our rules are right. As long as everything's within a box and a half. It's regular Michael V Hyland: where we get one or 2 that have 2 boxes or a little over 2 boxes. and again I'll take irregular, or I'll take slightly irregular.

0:38:32 Michael V Hyland: pretty flexible about that. The heart rate for this rhythm is between 150 and 250. Michael V Hyland: Normally. notice, there's always exceptions to exceptions to exceptions. I'll make the rules.

0:38:46 Michael V Hyland: This rhythm. although I have never seen it in 37 years of health care can go faster than 2 50 when it's and I will not give you this rhythm on homework or task. Michael V Hyland: But when it goes faster than 2 50, we call it ventricular flutter.

0:39:07 Michael V Hyland: This rhythm also may occasionally be less than 1 50, which I have seen several times. so when we get Vtac less than 1 50. We call it slow, Vtac.

0:39:23 Michael V Hyland: There's no P. Waves. Michael V Hyland: there's no pri, and then the Qrs Michael V Hyland: will be point 1 2 s or greater. Michael V Hyland: and T. And R usually go in the opposite direction. And here is a classic example of ventricular tachycardia.

0:39:44 Michael V Hyland: So Michael V Hyland: are you a little confused? Looking at this. Okay. would it help? You understand? If our wave is negative or positive?

0:40:02 Michael V Hyland: So so again, like I said, this is one PVC after another after another after another. Michael V Hyland: The waves at the bottom are more pointy. The waves at the top have a little bit of roundness, and then have a slope. so the T. Waves are up, the R. Waves are down.

0:40:27 Michael V Hyland: So when I'm looking for my regularity, I'm gonna count from the R wave to the R wave to the R. Wave to the R. Wave. and again it's going to be a fast rate. So you've got to be accurate with your small boxes. Michael V Hyland: So 4 and 3 make 7,

0:40:45 Michael V Hyland: and the next one is 7, and the next one is 7. Michael V Hyland: Actually, the second one might might be slightly less than 7. It might be 6 and 3 quarters. but we're going to call it 7. So if you go across the bottom of that, it looks like it's 7 all the way across. So this rhythm is regular.

0:41:10 Michael V Hyland: 7 small boxes in the 1,500. It's going to give me a heart rate of 214. Michael V Hyland: He waves Michael V Hyland: none. P. Ri. None. Michael V Hyland: Qrs. Now where are we measuring the Qrs. So I told you the tea wave is up. Michael V Hyland: See this bend right here. That's the end of the T-wave in the beginning of the Qrs. Right there. It's also right there. Michael V Hyland: It's right there, there, there! Michael V Hyland: Now, where does it end. You see the bend right here. You see that little bend. So we're going to measure right from here Michael V Hyland: right to that bend.

0:41:57 Michael V Hyland: one to 3, Michael V Hyland: one Michael V Hyland: oops. Michael V Hyland: 1, 2, 3. So the QRS. Is 3 small boxes Michael V Hyland: at point 1 2 s.

0:42:17 Michael V Hyland: so the rhythm's regular. Michael V Hyland: The QRS. Is wide without P. Waves. R is down. T is up. Our interpretation of this rhythm is ventricular. Tacha cardio. Michael V Hyland: What I want you to know about this rhythm is this is an immediate life, threatening rhythm. It's important that you quickly recognize life-threatening rhythms.

0:42:44 Michael V Hyland: This rhythm is life-threatening Michael V Hyland: for 2 reasons. Number one Vtac. We don't know if it profuses or not. I've already made that statement in class today. Michael V Hyland: Right? Number 2, Michael V Hyland: a heart rate of 214. Michael V Hyland: That's that's 3 and a half ventricular contractions Michael V Hyland: per second. Michael V Hyland: There's even if this rhythm was profusing. Michael V Hyland: there's no way blood is getting ejected, because how does it have any time to fill with blood in between contractions. So I'm going to guarantee you that if you felt for a pulse, this patient's not going to have a pulse

0:43:29 Michael V Hyland: right, if you're in the hospital somewhere, you better push the code blue button. If you're in a sleep lab somewhere, you better pick up the phone and dial 911, Michael V Hyland: because this patient's going to die. If you don't do anything, and you guys will all be Cpr. Certified right Michael V Hyland: before you go to a hospital for respiratory before you go to a sleep lab for Poly-some. So once you call 9 1 one, what else are we going to do? Michael V Hyland: We're going to start. Cpr. we're gonna get an Aed. We're gonna put the Aed on. And if it says shock, we're gonna shock. This is a shockable rhythm.

0:44:05 Michael V Hyland: Okay. Otherwise this patient's gonna die. Michael V Hyland: You should have a model. not on your target. Michael V Hyland: right? Nobody dies on my toxin.

0:44:19 Michael V Hyland: Such a headache, such a headache. All right questions on V-tech. Michael V Hyland: all right, moving on to the next. Rhythm. ventricular fibrillation. So in this rhythm the underlying causative agent. Here is we have multiple focal points Michael V Hyland: within the ventricles

0:44:44 Michael V Hyland: that are firing all at the same time. Michael V Hyland: so they create chaotic. Michael V Hyland: irritable impulses that the heart cannot follow.

0:44:59 Michael V Hyland: This is kind of like a fip right where we have multiple focal points in the atria and the atrium are trying to respond to every impulse, and they fibrillate rather than contract. This is exactly the same thing, except Michael V Hyland: it's doing it to the ventricles. Michael V Hyland: This is also an immediate, life-threatening rhythm, because Michael V Hyland: the ventricles are fibrillating. They are not pumping. There will not be a pulse. There will not be a blood pressure. Michael V Hyland: Need to pick up the phone. Call. 9, 1, one, push the code blue button, start Cpr. And get an Aed.

0:45:42 Michael V Hyland: So Michael V Hyland: the regularity. the word I want you to use for the regularity here is chaotic. Michael V Hyland: I'm gonna flash to the rhythm. Here's what an example of what Vfib looks like. Do you see any recognizable waveforms on that rhythm?

0:46:06 Michael V Hyland: So it is completely Michael V Hyland: irregular and crazy, looking so in terms of regularity. We're going to use the word chaotic.

0:46:20 Michael V Hyland: so the regularity is chaotic. There's no rate, no P. Waves, no pri, no Qrs. Michael V Hyland: Pretty recognizable rhythm. Michael V Hyland: and your interpretation is, gonna be be fair.

0:46:37 Michael V Hyland: And then we have 2 more ventricular rhythms to go through the next one Michael V Hyland: ventricular escape rhythm also known as idio ventricular. You can use either one Ver or idioventricular.

0:46:58 Michael V Hyland: and this is the last backup pacemaker that the heart has. So, in the absence of a higher pacemaker, IESA. Node a v. Node. Michael V Hyland: The ventricles initiate a regular impulse at the inherent rate. What's the ventricle's inherent rate? We learned this way in the beginning of the semester.

0:47:25 Michael V Hyland: 20 to 40 got to commit that to memory. Michael V Hyland: What's the inherent rate of that of the junction? Michael V Hyland: 40 to 60 right. But the junction has 3 different rates. Right? Has an 2 irritable rates in addition to the escape Michael V Hyland: accelerated 60 to a hundred, and then tachycardia when it's over a hundred. Okay? Michael V Hyland: So again, this is an escape mechanism. When the 2 hire pacemaker fails.

0:48:03 Michael V Hyland: So Michael V Hyland: it's usually pretty regular. The Q Rss. Are spread out pretty good. because the rate is so slow. So it's regular. The rate is usually between 20 to 40, Michael V Hyland: although it can start to drop below 20.

0:48:25 Michael V Hyland: If it drops below 20, it's generally a sign of a dying heart. We actually call that rhythm agonal. I won't give you a Ver. Less than 20, but Michael V Hyland: it's called diagonal. Again. There's no P. Waves, there's no Pri, and the Qrs will be point 1 2 s or wider, and generally R and T. Go in the opposite direction, and this is what it looks like. Michael V Hyland: So again, our waves are down. T. Waves are up. Michael V Hyland: So what we're going to do is count the spacing here. Michael V Hyland: 5, 1015, 2025, 30, 35, 40, 45, 50, 55, 57 and a half.

0:49:13 Michael V Hyland: 5, 1015, 2025, 30, 35, 40, Michael V Hyland: 45, 50, 55. So it's slightly irregular. Michael V Hyland: I would divide that into 56. I'm sorry I would take 56. Divide that into 1,500. Michael V Hyland: That heart rate's probably gonna be pretty close to 30, Michael V Hyland: 2726, 27 Michael V Hyland: P. Waves. None. Pr, I none. Michael V Hyland: Qrs, so we're going to measure from this up slope here, and we're looking for that bend right here as that bend. You can consistently see that bend all the way across.

0:50:02 Michael V Hyland: 1, 2, 3, 4, 5, 6, 7. So that is Q. Rs is 0 point 2 8. Michael V Hyland: Why do we? Divided by the payment limits slightly? Michael V Hyland: You're gonna come up with 27 or 30, right? Michael V Hyland: When it's slightly irregular, you could take the number of the average number of small boxes between. Michael V Hyland: and you're gonna come pretty close to whether, if you do it just out the R waves Michael V Hyland: right? So both will be pretty close. Michael V Hyland: and it'll certainly be within the 6 or 7 beat Grace, that I give you Michael V Hyland: alright. so you could say 30 or 27 alright. And then

0:50:52 Michael V Hyland: my interpretation is going to be Ver, or video ventricular. Michael V Hyland: This is also a life-threatening rhythm. A heart rate of 30 or 27 is not does not provide enough cardiac output to sustain life.

0:51:12 Michael V Hyland: So you're calling 9, 1 one right away still, gonna get the Aed because it might turn into? It might deteriorate in the Vfan

0:51:23 Michael V Hyland: questions on ideal ventricular. All right. The last one is called a cysteine. What does the prefix a mean? Michael V Hyland: Say.

0:51:41 Michael V Hyland: what is it? Michael V Hyland: It's without without the prefix a in the English language means without. Michael V Hyland: So without Sistoli.

0:51:54 Michael V Hyland: without contraction. Michael V Hyland: right? And that usually occurs when there's no electrical activity. Michael V Hyland: so the heart has lost its electrical activity. Michael V Hyland: There is no pacemaker to initiate electrical flow.

0:52:12 Michael V Hyland: so there is no regularity, no rate, no P. Waves. No pri, no qrs. And this is what it looks like. Michael V Hyland: This is, gonna be the hardest rhythm you interpret all semester long. Michael V Hyland: and if you miss this. If you miss this. you should check your pulse. Do you have a pulse? Michael V Hyland: I'm a funny guy, all right. Funny guy Michael V Hyland: questions on a cysteine.

0:52:43 Michael V Hyland: Alright. So agonal is a term used to describe a terminal lethal arrhythmia, and it's usually Michael V Hyland: Ver that slows down to a rate less than 20. Michael V Hyland: So when the heart is stopped beating in a reliable pattern. It's also called a dying heart or agonal. That's what an agonal rhythm is like. Agonal breathing is gasping. It's not really breathing.

0:53:10 Michael V Hyland: Okay. Now, there is one rhythm that I didn't really talk about. Michael V Hyland: Maybe I did. But is it possible to have mechanical activity without electrical activity. That should be an easy answer based on the rhythm that we just covered. Michael V Hyland: You're saying, yes, you can have mechanical contraction without a stimulus.

0:53:34 Michael V Hyland: No, you can't. Michael V Hyland: Can you have electrical activity without contraction? The answer to that is, yes. So we see a normal rhythm up on the Monitor. But my patient is blue and unresponsive. Michael V Hyland: and when I check for a pulse they don't have a pulse. Michael V Hyland: so that means they have a rhythm. Michael V Hyland: and that rhythm could be sinus, tachycardia, normal sinus, rhythm, sinus, bread, and cardiac.

0:54:06 Michael V Hyland: But when I check, they have no pulse. That's a code situation Michael V Hyland: that's a dying person, right? Michael V Hyland: And so you would need to call 9 1 1 start Cpr and get a defibrillator. We call that. And this is not gonna be on the homework because I have no way to do this Michael V Hyland: on a homework or a test, but that is anybody know what this rhythm is called

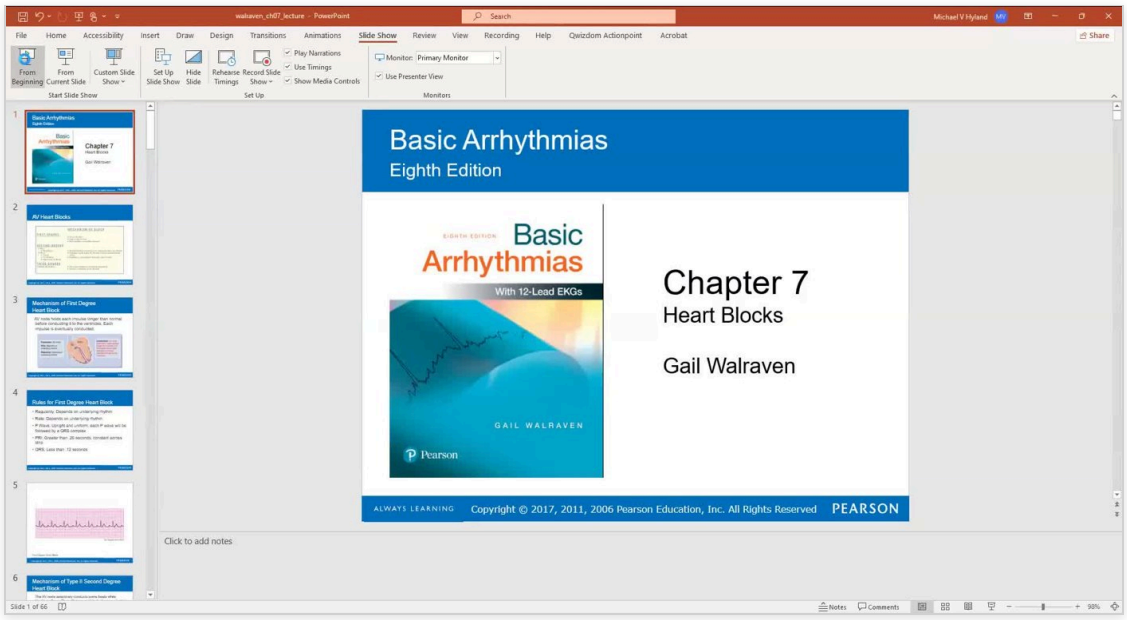
0:54:33 Michael V Hyland: PEA. Pulseless electrical activity. I think we did talk about right. I thought I said, I may have talked about right. Alright.

0:54:46 Michael V Hyland: How about we take a 4 min break because blocks are harder than what I just covered. Michael V Hyland: And and you know the homework is going to include ventricular and heart blots. Michael V Hyland: But next week we're going to review all of this stuff again.

0:55:12 Michael V Hyland: because this is a lot to take in in 1, 2 h lecture. Michael V Hyland: It's a no-house over. Michael V Hyland: Not so about you. Sorry. Michael V Hyland: Sorry be there. Michael V Hyland: we are progressing. Yeah. the video.

2 Chapter 7: Heart Blocks (title slide)

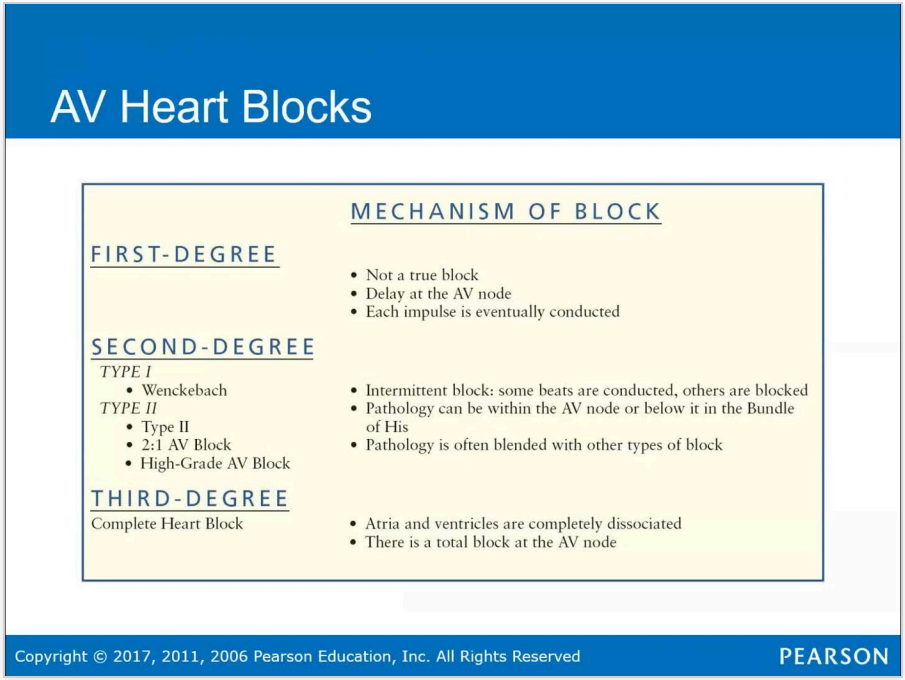
0:56:16 – 0:56:35



0:56:23 Michael V Hyland: Okay? So Michael V Hyland: again, we're now talking about the Michael V Hyland: probably the hardest rhythms to interpret.

3 AV Heart Blocks

0:56:35 – 1:06:20



0:56:37 Michael V Hyland: So there is multiple heart blocks that we're going to discuss first degree, heart block. Michael V Hyland: second degree, heart block, of which there is second degree, heart block, type, one Michael V Hyland: and second degree, hard block type 2,

0:56:55 Michael V Hyland: and then we have third degree heart block. Michael V Hyland: And we could actually have 2 different appearing third degree heart blocks depending on what's pacing the lower 2 chambers.

0:57:10 Michael V Hyland: Anyways, the problem here with heart blocks Michael V Hyland: is in the ab node. The AV node is malfunctioning. Michael V Hyland: so a couple of different things occur at the AV. No.

0:57:33 Michael V Hyland: So the essay node fires, and it sends that impulse along the intra atrial pathway and along the intranoto pathway. The AV node receives the impulse.

0:57:48 Michael V Hyland: and it holds on to the impulse for longer than it should. That's one problem Michael V Hyland: that the AV node holds on to the impulse before it sends it down Michael V Hyland: for ventricular contraction. I really should state that it holds on to the impulse longer than it normally does before it sends it down for ventricular contraction. Michael V Hyland: because the AV. No does hold the impulse for a very short period of time. but when it comes to heart blocks it holds it longer. then it should. How would we pick up on? If the A B node is holding on to the impulse longer than it should?

0:58:35 Michael V Hyland: What do we measure? What have we been measuring all semester long? That would tell us the Pr. I. Michael V Hyland: So when we see a pri longer than normal, what's normal? Pri

0:58:52 Michael V Hyland: point 1 2 to point 2 right or 3 to 5 small boxes. Michael V Hyland: So now, if we're starting to see a pri longer than 5 small boxes, that's an elongated abnormal pri. That generally means that the AV node is holding onto that impulse for longer than it should.

0:59:15 Michael V Hyland: The other problem that the A B node presents Michael V Hyland: is the essay node fires. Again. It sends that impulse in the intranodal pathway. Michael V Hyland: The A B node catches the impulse Michael V Hyland: and doesn't send it down for ventricular contraction.

0:59:36 Michael V Hyland: How are we gonna pick up on that Michael V Hyland: on an ekg. Michael V Hyland: so you're right. There'll be no Qrs.

0:59:53 Michael V Hyland: What will there be not followed by a Qrs. Michael V Hyland: there's gonna be A, Michael V Hyland: are you? All right over there? Michael V Hyland: We did it. Thank you.

1:00:13 Okay. Michael V Hyland: spin. Michael V Hyland: Gonna sit up. Video. So Michael V Hyland: then I got me to go Michael V Hyland: show them. I just don't. Wanna, you know, Spread, because

1:00:32 Michael V Hyland: me to bring outside Michael V Hyland: we don't like, and things want to get out. But I'm afraid so. Michael V Hyland: It's really Michael V Hyland: thank you, though proudly. Well. Michael V Hyland: another question

1:00:58 Michael V Hyland: help me. Michael V Hyland: So what will we see on the Ekg strip? Remember the sa node fires. It sends the signal along the Michael V Hyland: intra atrial pathway. Why does it do that?

1:01:38 Michael V Hyland: And tra atrial pathway? Not difficult people? Michael V Hyland: Why does it send the signal to the atrial cells Michael V Hyland: or atrial contraction.

1:02:00 Michael V Hyland: you know atrial contraction. Remember, we talked about the hearts meant to beat from the top down. Michael V Hyland: We want the atri to maximize the blood content in the ventricles before the ventricles contract. What waveform does the atrial contraction form? He way? So we're gonna see a P wave not followed by Michael V Hyland: a QRS. So blocks heart blocks are gonna create 2 things longer, pris Michael V Hyland: and blocked beats. How do we recognize a blocked beat on a on a on an Ekg strip. Michael V Hyland: Excuse me. Michael V Hyland: A pea wave, not followed by a QRS.

1:02:45 Michael V Hyland: Okay. understand those 2 concepts. Those are important concepts. Michael V Hyland: In order to interpret hard blocks correctly. Michael V Hyland: The other thing that I will say is

1:03:06 Michael V Hyland: that's really important Michael V Hyland: is making sure you do a good job of measuring the Pr. I. Michael V Hyland: Because the other telltale sign that enables you to more easily interpret heart blocks is getting an accurate measurement of the pri. Michael V Hyland: because the Pr. I

1:03:33 Michael V Hyland: is really a key to interpret hard blocks. Michael V Hyland: You'll see what I mean as we go through. So the first rhythm we're gonna talk about is first Degree heart block. It's not a true block, it's just a delay. Michael V Hyland: So in this particular rhythm Michael V Hyland: the ab node holds on to the, to the impulse longer than it should. However, every impulse is sent down for ventricular contraction. So in first degree, Heart Block, it's not a true block, because every beat is conducted eventually.

1:04:15 Michael V Hyland: so every p-wave will be followed by a Qrs. Because none of the beats are blocked. Michael V Hyland: and we're going to go over this again when we when I pull up the rules, and I'll pull up the rhythm and we'll go through the data collection for these

1:04:33 Michael V Hyland: second degree hard block. There's 2 types. There's one where we're going to get a lengthening Pr interval Michael V Hyland: for 2 or 3 or 4 beats. followed by a blocked beat. and then the pattern's going to repeat itself again.

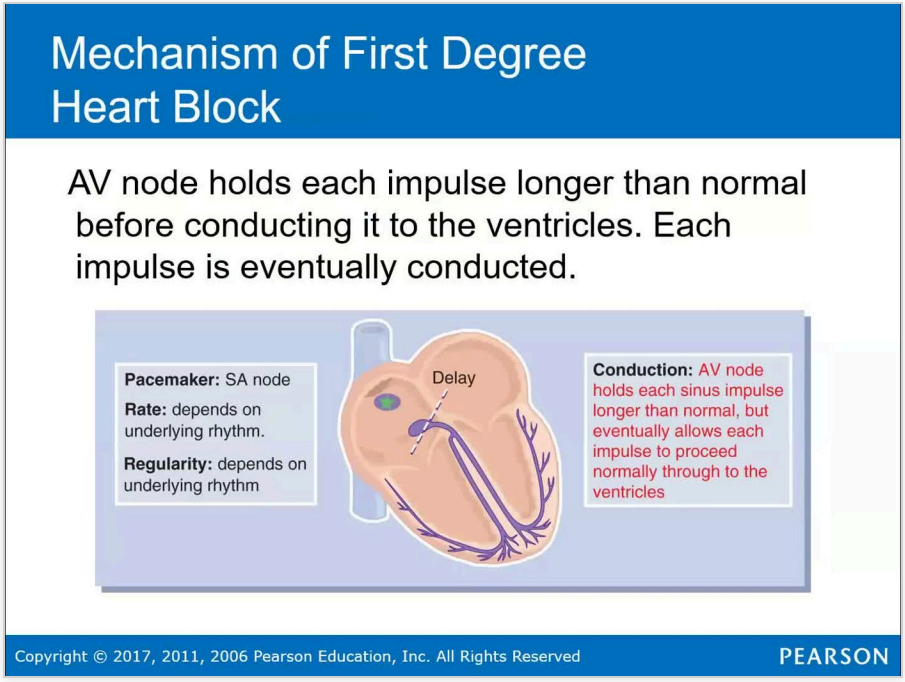
1:04:51 Michael V Hyland: Again, we'll go over that second degree. Type 2. Michael V Hyland: We're going to see P. Waves that are blocked. But we're gonna see other P. Waves that are not blocked. Michael V Hyland: and the key there is going to be the P. Waves that are conducted, IE. The P. Waves that are followed by a QRS. That PRI will stay the same all the way across the strip. Michael V Hyland: So in second degree, type one, the Pr. Interval gets longer, longer, longer, longer, until we have a blocked beat. Second degree. Type 2. We have some beats that are blocked, some beats that are conducted. The beats that are conducted will have the same pri all the way across the strip. So when you see P waves that are blocked, but the P wave is the same Michael V Hyland: second degree. Hard block type 2, Michael V Hyland: and then third degree block. The AV node is completely blocked. Nothing Michael V Hyland: from any of the super ventricular like the essay. No, to the atria. Those impulses cannot get through the Ab. No. So we're going to have atrial activity.

1:05:56 Michael V Hyland: and atrial activity is doing its own thing because none of the impulses actually get through the A. V note. So therefore, another pacemaker has to pace the lower 2 chambers. Michael V Hyland: Otherwise the patient's gonna die. You won't have any

ventricular heartbeats right? Michael V Hyland: And it can be paced by the lower portion of the junction or the ventricles themselves. All right, let's get into this.

4 Mechanism of First Degree Heart Block

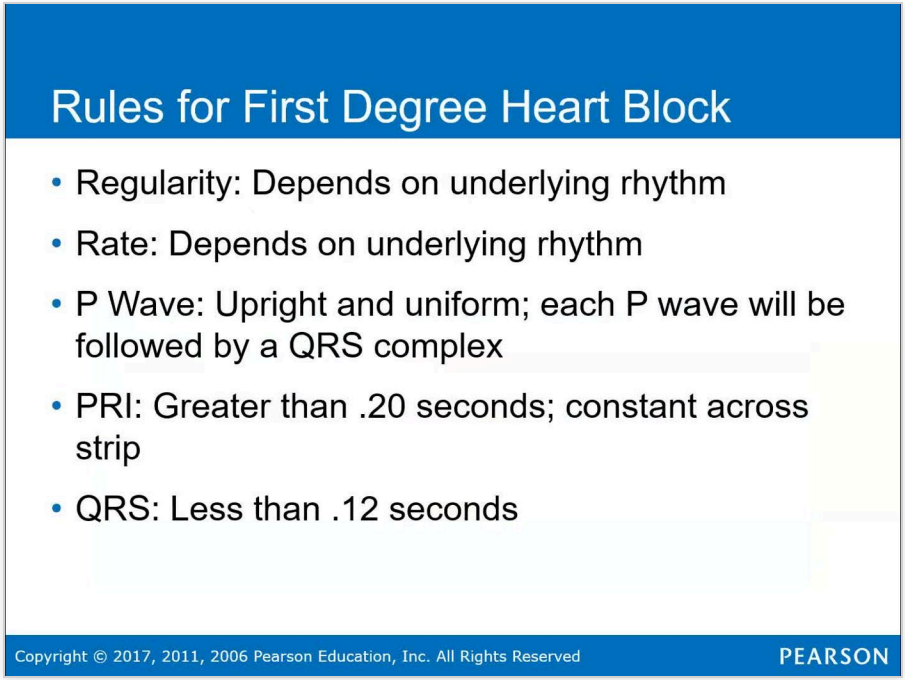
1:06:20 – 1:06:51



1:06:22 Michael V Hyland: So mechanism of first degree art block. So the AV node holds each impulse longer than normal before conducting it to the ventricles. Michael V Hyland: Each impulse is Michael V Hyland: conducted so that means there's no blocked beats here. When I say the impulse is conducted. That means it's sent down for ventricular contraction or ventricular depolarization. Michael V Hyland: Okay, so let's look at the rules

5 Rules for First Degree Heart Block

1:06:51 – 1:07:53

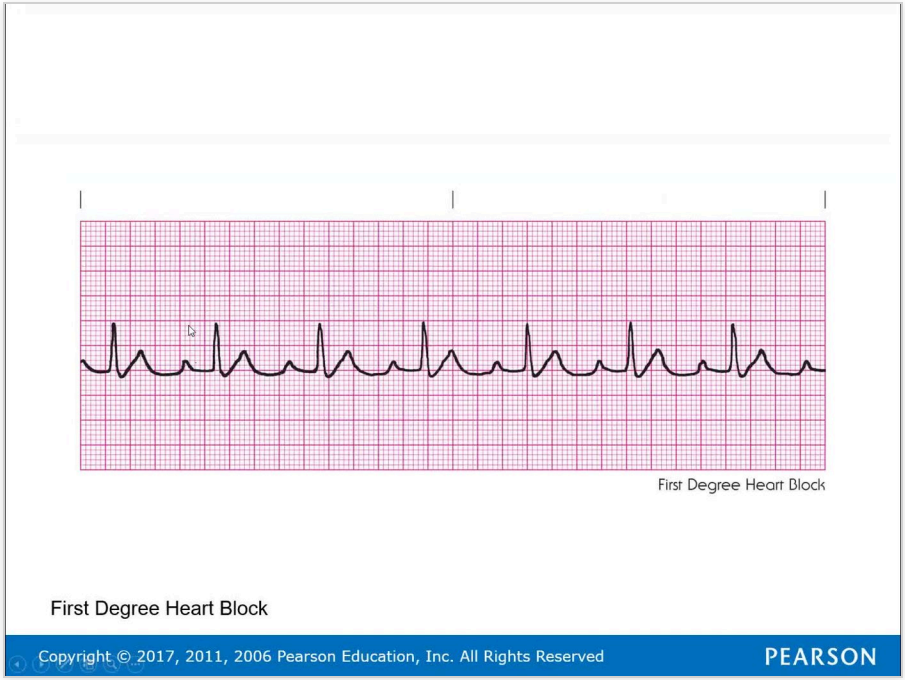


1:06:52 Michael V Hyland: the rules for this the rhythm is usually regular. Michael V Hyland: and in this case we're gonna have to interpret the underlying rhythm Michael V Hyland: and the first degree heart block.

1:07:08 Michael V Hyland: The rate is going to depend on the underlying rhythm. The P. Waves will be upright and uniform. Each P. Wave will be followed by a QRS. Michael V Hyland: So therefore, we will be able to measure the pri. But the pri here is going to be point 2 s. Michael V Hyland: or greater, usually greater than point 2 s point 2 is normal, so it would have to be point 2 one or point 2 2 to have a first degree block. Michael V Hyland: and that PRI will be the same all the way across the strip, so every PRI will be point 2 2 going across the strip. and the QRS. Will be narrow. Michael V Hyland: So let's take a look at it.

6 First Degree Heart Block (ECG strip)

1:07:53 – 1:12:00



1:07:55 Michael V Hyland: So let's gather our information. Michael V Hyland: 5, 1015, (161) 718-1920. Michael V Hyland: What do you call it 20 and a half or 21, Michael V Hyland: 5, 1015, 18, Michael V Hyland: and 3, we'll call it 215-10-1519, and 2 make 215-10-1520, Michael V Hyland: 21. The next one is 21. So this rhythm is regular.

1:08:33 Michael V Hyland: gives us a heart rate of 71 p. Waves. Now, you guys can see the P. Waves. They're right here. They're upright and they're uniform.

1:08:46 Michael V Hyland: Now we got to measure the Pr. I. Michael V Hyland: And just as a quick heads up. I know that this is a long pri. This is, if I just look at the last pri that I can measure on the strip all the way over here to the right. Michael V Hyland: This P wave starts before the large box Michael V Hyland: and the Qrs. Occurs after the large box. So I know that this Pri is longer than one large box. So there's definitely a delay here. Right? So let's actually measure.

1:09:19 Michael V Hyland: And I want Michael V Hyland: 2, 3, 4, 5, 6. I think this P wage starts halfway in between and that Qrs starts halfway in between. That's 6 small boxes.

1:09:34 Michael V Hyland: That's 5 6. The next one might be 6 and a half. Michael V Hyland: 1, 2, 3, 4, 5, 6 Michael V Hyland: and a half.

1:09:51 Michael V Hyland: 1, 2, 3, 4, 5, 6 and a half. All right. So we'll call it 6 and a half. That's going to be Michael V Hyland: point 2 6.

1:10:06 Michael V Hyland: You RS, Michael V Hyland: 2 small boxes. Michael V Hyland: 1, 2, yeah, 2 small boxes at point 0 8. So everything in this rhythm is normal except for the pri. So what's the underlying rhythm?

1:10:29 Michael V Hyland: Normal sinus rhythm with first degree heart block. Michael V Hyland: So first degree harp is pretty simple. You see that every Qrs is followed or every p-wave is followed by a Qrs.

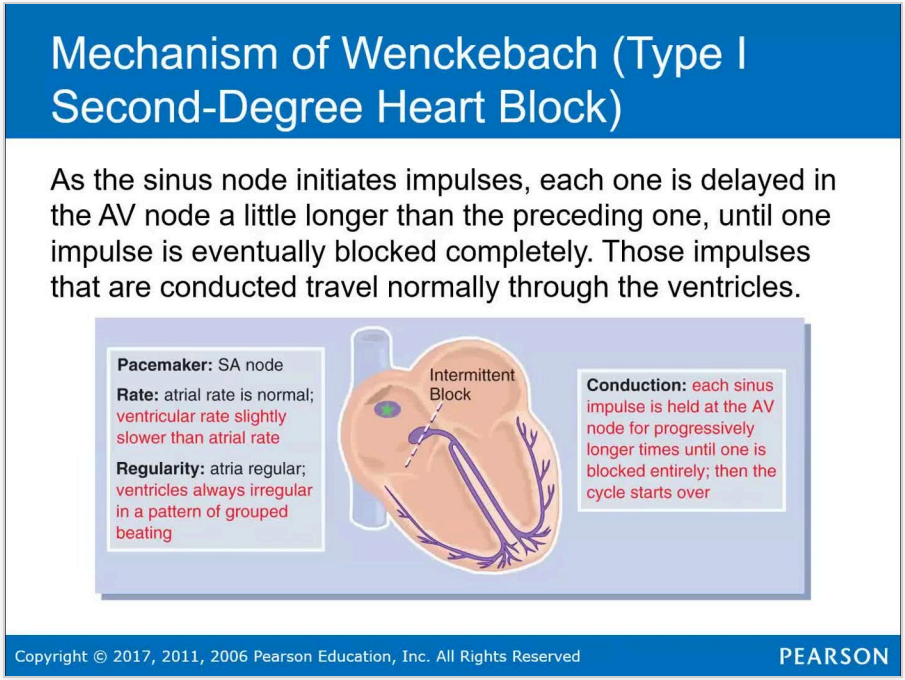
1:10:48 Michael V Hyland: It's just an elongated pri Michael V Hyland: question. Can we use one in the Degree school, or do we have to write out first? Michael V Hyland: How about if you go? One with the degree symbol followed by HB.

1:11:09 Michael V Hyland: You could go. Nsr, nsr. Michael V Hyland: the letter W. Slash one degree HB. Michael V Hyland: I mean, I need to save your hand from writing for sure that you're killing already.

1:11:30 Michael V Hyland: At least, I'm not killing your hand. I'm not. It's not a 2 for Michael V Hyland: questions on first degree. Harpal. Alright. Michael V Hyland: Again, the book goes into second degree. Heart block type Michael V Hyland: 2. So I'm gonna come back to this. I'm gonna go to second degree heart block type one. And then I'll come back to that

7 Mechanism of Wenckebach (Type I Second-Degree Heart Block)

1:12:00 – 1:14:33



1:12:01 Michael V Hyland: so second degree Heart Block type one Michael V Hyland: also known as Wankie Bach. Michael V Hyland: also known as Mobits, one MOB.

1:12:25 Michael V Hyland: ITZ. Mo. Bitz one. and the one is Roman numeral one. Michael V Hyland: So it's Michael V Hyland: the letter. It's a capital I, Roman numeral one

1:12:41 Michael V Hyland: so second degree heart block type one Wankybach or mobits one. Michael V Hyland: So what's happening? Here is again the sinus note is the pacemaker. As the A V node receives the impulses. each one is delayed in the A V node a little bit longer than the preceding one. Michael V Hyland: Well, how do I pick up on that Michael V Hyland: is, have a Pr. I. Michael V Hyland: Now my next pri is going to be longer, because if it holds onto it, it's going to elongate the pri. Michael V Hyland: Does that make sense and will make sense more when we see the rhythm Michael V Hyland: so? And it might do this Michael V Hyland: for 2 beats, 3 beats or 4 beats.

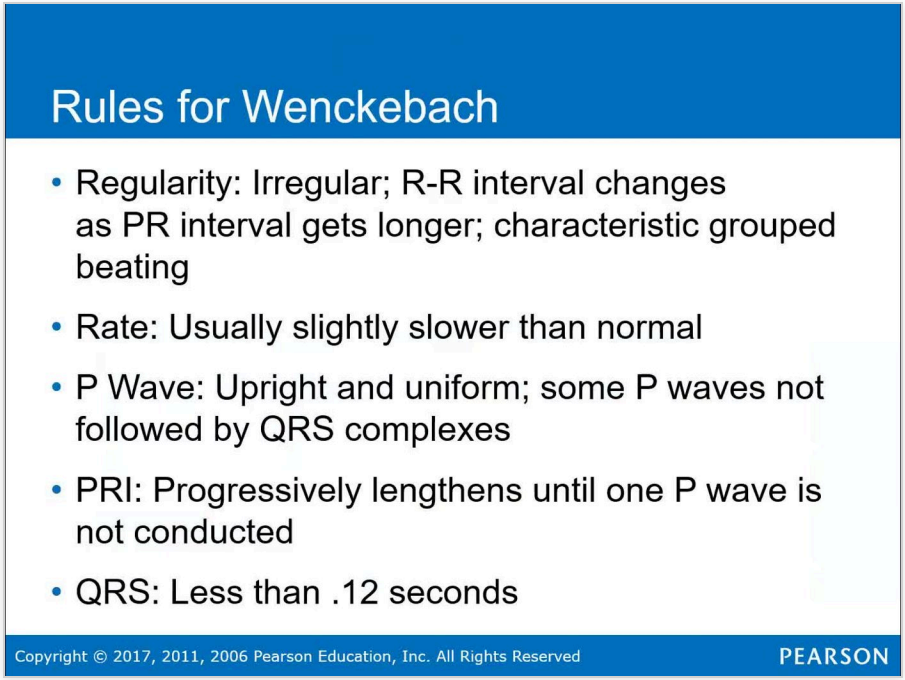
1:13:31 Michael V Hyland: You're going to see a lengthening Pr. I. Michael V Hyland: Until it blocks one beat completely. Michael V Hyland: and then you'll see a P. Wave not followed by a QRS.

1:13:46 Michael V Hyland: So the key or the trick. Here is a pri that gets progressively longer until I see a p-wave, not followed by a Qrs. Michael V Hyland: and then Michael V Hyland: the pattern repeats itself over.

1:14:04 Michael V Hyland: So AV node holds on to it a little bit longer than the preceding one until an impulse is eventually blocked. Michael V Hyland: The impulses that are conducted will have a narrow QRS. So let's look at the rules. So what this is is an intermittent block. Michael V Hyland: It lets 2 or 3 beats go through, and then it blocks a beat. It lets 2 or 3 beats go through, and then it blocks a beat.

8 Rules for Wenckebach

1:14:33 – 1:16:22



1:14:34 Michael V Hyland: So this rhythm is always irregular. Michael V Hyland: In fact, there's actually a special name for this. and that is Michael V Hyland: regularly irregular.

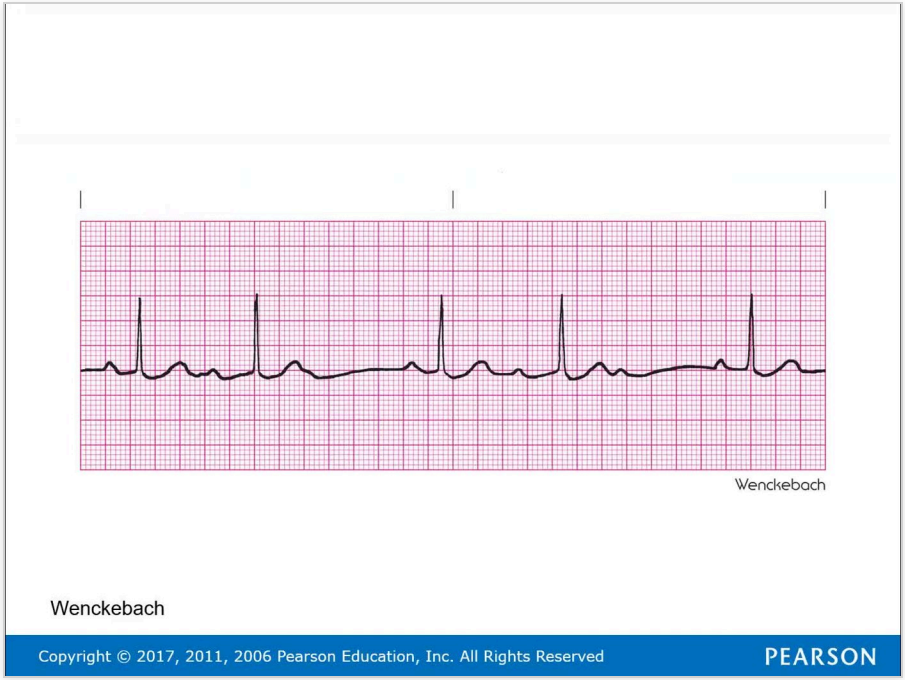
1:14:54 Michael V Hyland: I will take regularly irregular or irregular. Michael V Hyland: what does regularly irregular mean? Michael V Hyland: So the rhythm itself is irregular, but there's a pattern to the irregularity.

1:15:12 Michael V Hyland: It's good to know you got that in here Michael V Hyland: regularly irregular Michael V Hyland: fits. Your personality? Yeah, all right. So it's going to have an irregular R to R. It has to have an irregular R to R. The Pr. Interval lengthens. Michael V Hyland: because when the Pr interval lengthens you're pushing the R. Wave out farther. so there's more distance, and that distance grows as you push it out farther and farther.

1:15:44 Michael V Hyland: and the other thing that sometimes this gets characterized by is grouped beats. We have a group of 3 or 4 beats. Michael V Hyland: and then a blocked beat, and then we have a group of 3 or 4 beats, and then a blocked beat. So you'll see what I mean when I call up the rhythm. Michael V Hyland: the rate's gonna be a little bit lower, because we're blocking every third or fourth beat. P. Waves will be upright in uniform. Some peas will not be followed by a QRS. Those are the beats that get blocked. Michael V Hyland: The PRI. Is going to progressively lengthen until one P wave is blocked Michael V Hyland: and the Qrs will be normal or narrow.

9 Wenckebach (ECG strip)

1:16:22 – 1:23:22



1:16:22 Michael V Hyland: And let's look. let's look at this rhythm. Now Michael V Hyland: let me orient you to this rhythm, because it's a little tricky, at least in the beginning.

1:16:36 Michael V Hyland: So we see a pea wave. followed by a QRS. Followed by a t-wave. followed by a P. Wave Michael V Hyland: with an increased length. In the Pr. Interval you could see how much farther this P. Wave is from that QRS. Followed by a T. Wave. Michael V Hyland: What's tricky here is there is a P. Wave on the down slope of this T. Wave. Michael V Hyland: Does everybody see that right there. Michael V Hyland: So that is a P wave not followed by a Qrs.

1:17:13 Michael V Hyland: It's easier to see in the second grouping than it is. The first grouping Michael V Hyland: P wave, followed by QRST. Wave, P. Wave with an increased length of the pri, followed by a T-way P. Q Rs T. Wave, and then a P. Wave, not followed by a Q. Rs that beat got blocked. So this beat got blocked just like that beat got blocked. Michael V Hyland: and you see the grouped beatings. Here we have a group of 2, a group of 2, and if we could see this next beat we would have another group of 2.

1:17:47 Michael V Hyland: Let's gather information. Michael V Hyland: Put this all together. Michael V Hyland: 5, 1015, 20, and 3 makes 23 and a half.

1:18:02 Michael V Hyland: 51015202530, Michael V Hyland: 31, 32, 4, more, make 36, and 2 halves make 37, 5, 1015, 20, Michael V Hyland: 2223, 24, and a half. Michael V Hyland: 5, 1015, 2025, 30, 35, Michael V Hyland: 38. So this rhythm is irregular.

1:18:35 Michael V Hyland: When you look at it. Michael V Hyland: you see a group of 2 with a blocked beat, a group of 2 with a blocked beat, and the rhythm is irregular, but the pattern repeats itself, and that's what I mean by regularly irregular. Michael V Hyland: If you're confused about that, then just put irregular.

1:18:57 Michael V Hyland: Everybody good with that. Michael V Hyland: Okay? Hmm. Michael V Hyland: The rhythm is irregular, even if you put regularly irregular. we still have to do the irregular heart rate measurement here. There's 5 QRS. On the screen, so the heart rate is 50

1:19:16 Michael V Hyland: P. Waves are upright and uniform. Michael V Hyland: Can't see this one all the way, but it's going to be. If we could, it would look like that. Now we've got to measure the Pri. So this first pr, I

1:19:33 Michael V Hyland: is one Michael V Hyland: plus 5 and a half. So this first Pri is 6 and a half boxes. You see, it starts right on this line right here. Michael V Hyland: right there. Michael V Hyland: and it spans this whole entire large box. So there's 5 plus one, and then it starts halfway in between 6 and a half. Michael V Hyland: So this Pr, I is point 2 6.

1:20:01 Michael V Hyland: You might want to actually write that on the strip. Let's measure the next one. I know this one's gonna be longer.

1:20:13 Michael V Hyland: So it starts halfway in between. So if I go halfway in between here to halfway in between here, that would be 5 boxes. and I still got to keep count Michael V Hyland: 6, 7, 8, 9, 9 small boxes that Pr, I. Is point 3 6

1:20:40 Michael V Hyland: going to come over here and measure these. That's 3 Michael V Hyland: and 5. Make 8 point 3 2. Michael V Hyland: That's 3 and 5. Make 8 and one more makes 9,

1:21:00 Michael V Hyland: 4 and 2 and a half. Make 6 and a half. So what am I expecting you to do for the Pr. I here? I want you to give me a range. Michael V Hyland: You're going to be giving me 2 numbers, the shortest pri and the longest pri. Michael V Hyland: So the shortest pri. Here is 6 and a half small boxes, so point 2 6 Michael V Hyland: dash to the longest pri, which is point 3 6. Everybody understand what I'm expecting.

1:21:35 Michael V Hyland: Okay, we're going to measure the QRS. It looks like a box and a half Michael V Hyland: point 0 6. So what am I seeing here? I'm seeing a long P. Ri. so that must mean it's a block. and then I'm seeing it, followed by a longer Pr. I. Michael V Hyland: Followed by a blocked beat. Michael V Hyland: and then I see a long PRI. Followed by a

longer PRI. Followed by a blocked beat. Michael V Hyland: So when I see a lengthening pri followed by a block beat, it has to be Wankybach Michael V Hyland: or second degree heart block type one.

1:22:19 Michael V Hyland: So think to yourself, Winky Bach. lengthening pri with blocked beats. with a pattern that repeats itself over and over and over again

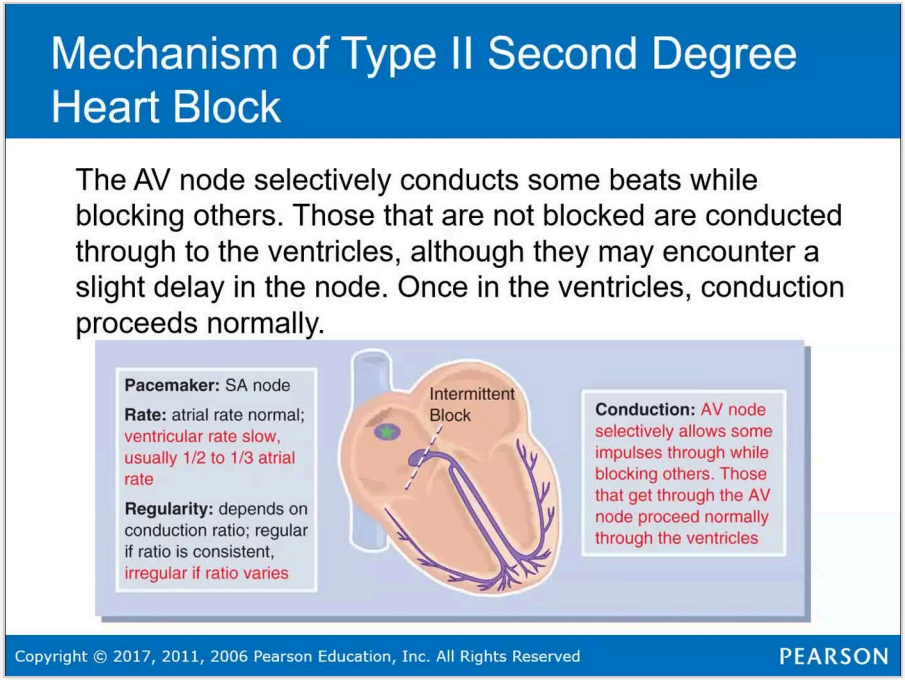
1:22:31 Michael V Hyland: like a broken record. If you've ever heard that expression. Michael V Hyland: some of you may not even know what a record is. Michael V Hyland: You know what a record is, you know

1:22:44 Michael V Hyland: I'm glad makes me feel a little less old Michael V Hyland: questions on Winkie Bach. You guys know what a record is over there? You sure? What's it made of?

1:23:06 Michael V Hyland: Okay. Michael V Hyland: you look pretty young. that's pretty young. Hello! Michael V Hyland: Alright! I believe you. Michael V Hyland: I believe you alright. So now we're gonna go back to second degree. Hard block type, 2

10 Mechanism of Type II Second Degree Heart Block

1:23:22 – 1:24:58



1:23:23 Michael V Hyland: mechanism of type, 2 s degree heart block. also known as Michael V Hyland: Classic heart block. the word classic heart block.

1:23:39 Michael V Hyland: also known as mobits. Michael V Hyland: 2 Roman numeral, 2, Michael V Hyland: the A. V node selectively conducts some beats while blocking others.

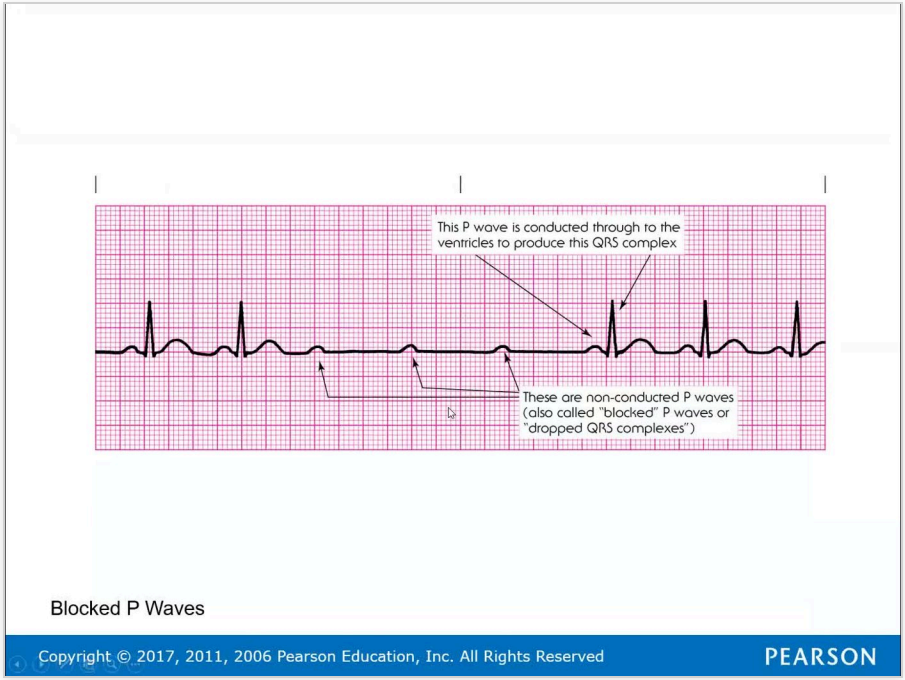
1:23:59 Michael V Hyland: Those that are not blocked are conducted through to the ventricle. Michael V Hyland: and they may encounter a slight delay in the A B known. So the beats that are conducted

1:24:13 Michael V Hyland: may have a normal P. Ri, or they may have a long P. Ri. so those that are not blocked are conducted through the ventricles, but they might encounter a delay which anytime we have a delay.

1:24:30 Michael V Hyland: We're going to have a long pri. Michael V Hyland: so this can occur with a delay or without a delay. Michael V Hyland: It's not necessarily important that you memorize whether there's a delay or not. Michael V Hyland: So I'm gonna give you the the trick to interpret this rhythm. Once in the ventricle conduction proceeds normally, that means the Qrs will be narrowed.

11 Blocked P Waves

1:24:58 – 1:29:10



1:24:59 Michael V Hyland: So we look at this first example. we're looking at the first beat, which has a P wave Michael V Hyland: and a normal pri, because it's less than one large box, followed by a narrow QRST. Wave. The next beat looks exactly the same. Michael V Hyland: and now I see a P. Wave not followed by a QRS. So this beat got blocked. Michael V Hyland: I see another P wave, not followed by a QRS. That beat got blocked. and then I happened to see a third P. Wave in a row, not followed by a QRS. That beat got blocked. Michael V Hyland: So when I see a P. Wave not followed by a QRS. It has to be a block.

1:25:45 Michael V Hyland: Okay, as I move to the latter portion of the rhythm here now, everything looks normal like it did in the beginning. Michael V Hyland: Harry.

1:25:56 Michael V Hyland: So we're gonna go through our information gathering Michael V Hyland: 5, 1015, Michael V Hyland: 1617, 1819, Michael V Hyland: 5, 1015, 3 and a half and a half make 19. And this is 19.

1:26:21 Michael V Hyland: So this underlying rhythm is regular. Okay. Michael V Hyland: not that we have to interpret the underlying rhythm for second degree heart block. We just call it second degree heart block. Michael V Hyland: and then P. Waves are upright and uniform. Michael V Hyland: and just some of them are blocked.

1:26:44 Michael V Hyland: This Pr. I. Michael V Hyland: Is 4 and a half small boxes, and it's consistently 4 and a half small boxes.

1:26:56 Michael V Hyland: So this rhythm is regular with blocked beaks Michael V Hyland: heart rate. Michael V Hyland: This is where the heart rate gets tricky.

1:27:18 Michael V Hyland: The atrial rate is 70, Michael V Hyland: the ventricular rate is not 70. We have 3 block beats in the middle of this rhythm. Michael V Hyland: so the ventricular rate is not 70.

1:27:35 Michael V Hyland: So because we have 3 block beats there, this ventricular rate is going to be maybe a third lower than the atrial rate.

1:27:47 Michael V Hyland: So I would just say, I see 5 QRS. On there. Michael V Hyland: My heart rate is going to be 50.

1:28:00 Michael V Hyland: The practice of underlying feats regularly treat it like they're Michael V Hyland: correct. Yeah, yes. Michael V Hyland: because it has blocked beats

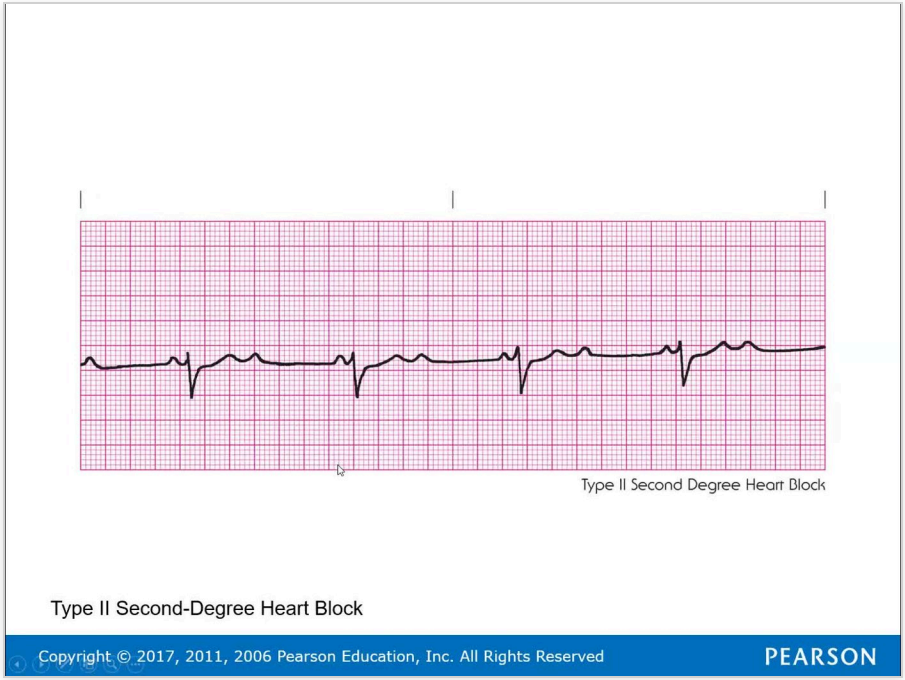
1:28:12 Michael V Hyland: because we can't. We can't rely on that rule because we don't know how many beats are going to get blocked by the A B node. So you're just going to count the number of R waves.

1:28:25 Michael V Hyland: This Qrs is pretty narrow, right? And the key here, the key. Here 2 keys I see blocked bees. Michael V Hyland: I see P. Wage not followed by a QRSI have blocked beats. Michael V Hyland: but the PRI stays the same. It's not like second degree harp block type one where the PRI. Lengthens lengthens, and then I have a block P. This PRI. Stays the same Michael V Hyland: for every beat that is conducted. Michael V Hyland: So when you see blocked beats with a pri that stays the same. Michael V Hyland: it's second degree heart block type 2 all day long. all week long.

1:29:06 Michael V Hyland: Questions about that. and this can. This can appear in a lot of different ways. Here's another example.

12 Type II Second-Degree Heart Block (ECG strip)

1:29:10 – 1:31:04



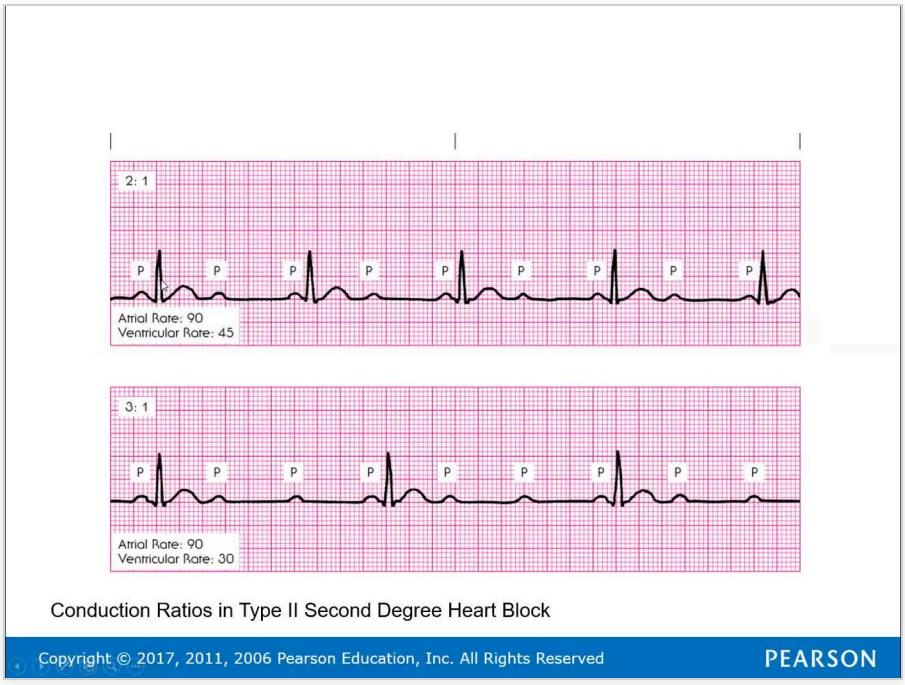
1:29:17 Michael V Hyland: So here we have a P. Wave that's blocked. followed by another P wave Michael V Hyland: that is conducted. It's followed by a QRST. Wave. Another P wave that's blocked. followed by another P. Wave with a QRST. Wave. Michael V Hyland: So what the AV. Note is doing here? Michael V Hyland: It is blocking every other beak, blocked, conducted, blocked, conducted. Michael V Hyland: Now because this block pattern is regular. Michael V Hyland: Now we can use our regular method to count the heart rate

1:29:57 Michael V Hyland: because this block is consistent. This is a 2 to one block. Or you guys understand what I mean by that. Michael V Hyland: for every 2 P. Waves I'm conducting one QRS. Michael V Hyland: 2 to one ratio. When the block ratio stays the same, the rhythm will be regular Michael V Hyland: when the block ratio varies, the rhythm will be irregular, and that's gonna be the difference of how you count the heart rate when the block is consistent, the ratio stays consistent. You can count the number of small boxes between our waves divided into 1,500. That's gonna be your heart rate Michael V Hyland: when the block varies, and it's not consistent like the last rhythm. Count the number of R waves and multiply by 10.

1:30:48 Michael V Hyland: So this pri is going to be consistent right? So we have blocked beats with a pri. That's the same all the way across the rhythm. It has to be second degree heart block, type. 2 Michael V Hyland: questions on how I got there.

13 Conduction Ratios in Type II Second Degree Heart Block (2:1, 3:1)

1:31:04 – 1:31:57

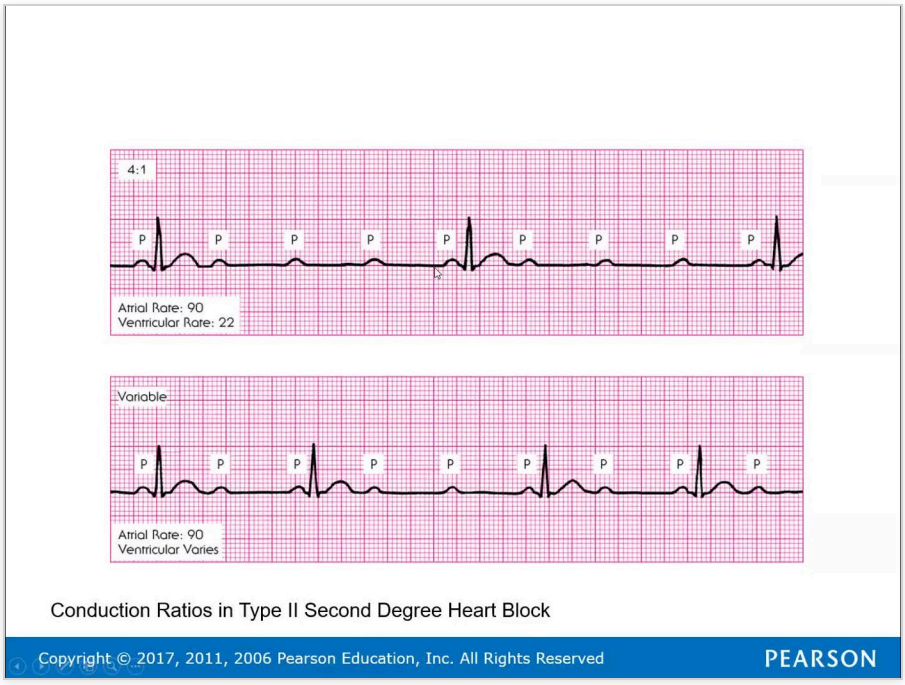


1:31:05 Michael V Hyland: So here are some examples, and this is the 2 to one ratio that I was talking about, blocked. Michael V Hyland: conducted, blocked, conducted. That's a 2 to one ratio. Michael V Hyland: The second degree heart block can appear in a number of different ways. Michael V Hyland: This is a 3 to one ratio, blocked, blocked, conducted, blocked, blocked, conducted. Michael V Hyland: When the ratio stays the same, the r spacing will be regular. so you can count your heart rate that way.

1:31:39 Michael V Hyland: That's a pretty slow heart rate. By the way, second Degree heart block. When the heart rate gets below 60 is a life-threatening rhythm. Need to pick up the phone and call 9, 1, one Michael V Hyland: or call a rapid response.

14 Conduction Ratios in Type II Second Degree Heart Block (4:1, variable)

1:31:57 – 1:33:15



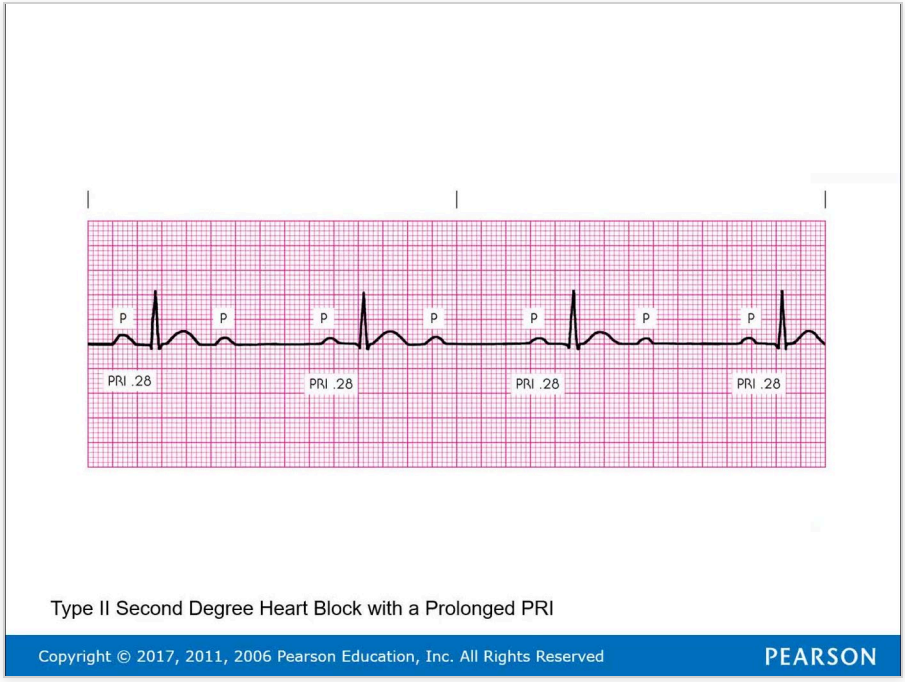
1:31:58 Michael V Hyland: It can even go 4 to one so blocked. Michael V Hyland: blocked, blacks conducted. Michael V Hyland: and this gives me a ventricular rate of 22, Michael V Hyland: and then the variable block like I was talking about. We blocked one, conducted one Michael V Hyland: blocked, one blocked one, conducted one. Michael V Hyland: blocked one, conducted one, so we went from 2 to one to 3 to one to 2 to one. This is going to make the R to our spacing irregular. Michael V Hyland: So this rhythm up here is regular. This rhythm is irregular. Michael V Hyland: So I'm gonna go 1, 2, 3, Michael V Hyland: 4, and that's my heart rate 40

1:32:43 Michael V Hyland: as opposed to Michael V Hyland: 5, 1015, 2025, 30, 35, 40, 45, 50, 55, 60, 65, 67, and a half divided into 1,500. That's gonna be a heart rate in the low 20 s. Michael V Hyland: Everybody understand the difference. I know you're kind of getting burnt at the end here.

1:33:05 Michael V Hyland: So if you look, these, all of these examples also come with a normal pri less than point 2 s.

15 Type II Second Degree Heart Block with a Prolonged PRI

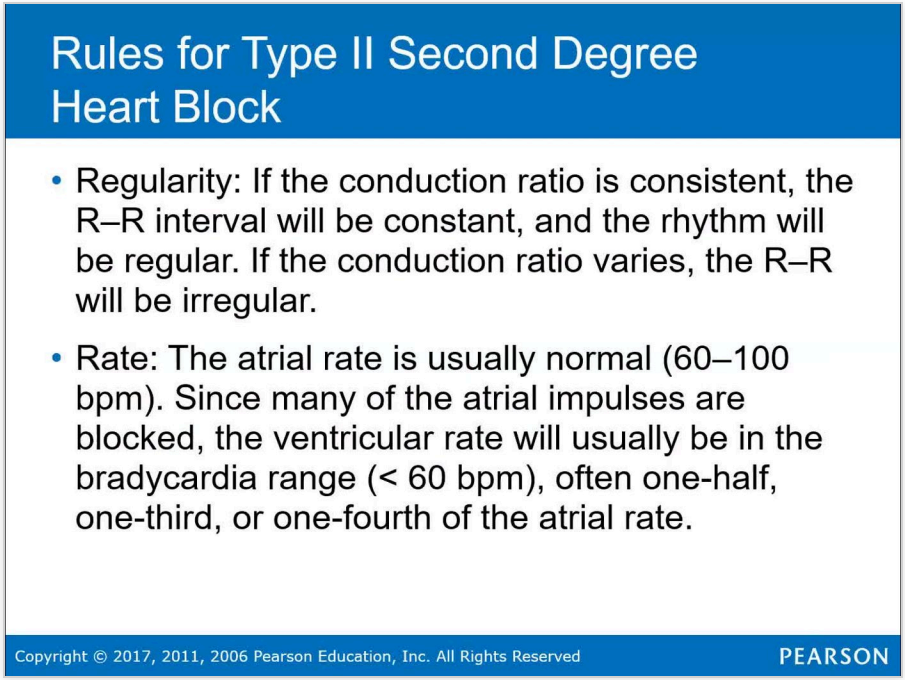
1:33:15 – 1:33:37



1:33:15 Michael V Hyland: I can show you this rhythm with a 2 to one block ratio, and the Pri is elongated. So second degree heart block type 2 can have a normal pri or a long pri, but the trick is, the pri stays the same all the way across the strip.

16 Rules for Type II Second Degree Heart Block

1:33:37 – 1:34:23

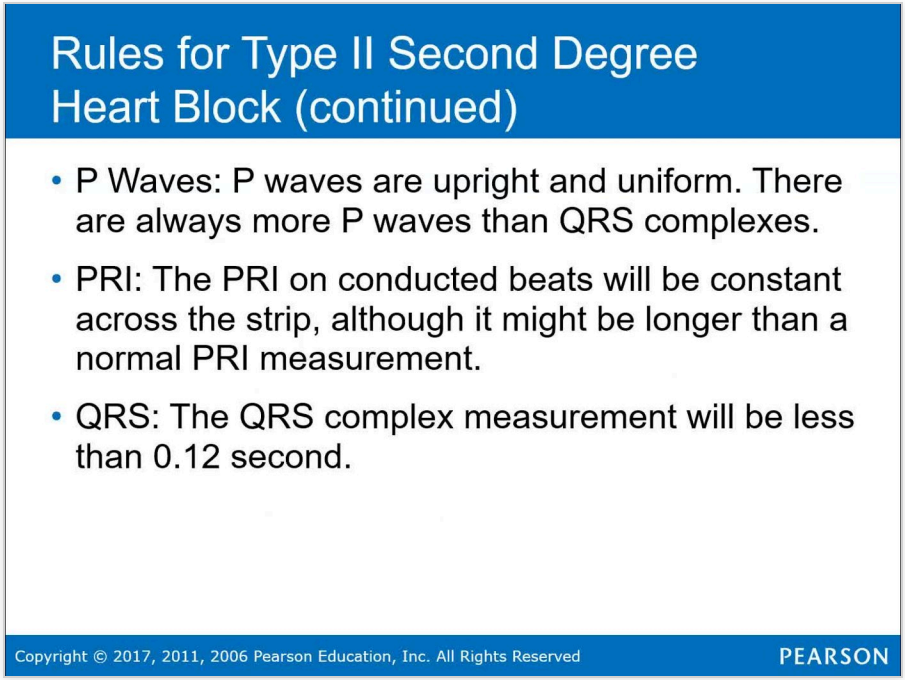


1:33:38 Michael V Hyland: So regularity, if the conduction ratio is consistent. Michael V Hyland: Another way of saying, that is, if the block ratio is consistent, R to R will be constant, the rhythm will be regular. Michael V Hyland: If the conduction ratio varies. R to R will be irregular.

1:34:00 Michael V Hyland: If R to R is regular, take the spacing divided into 1,500 for the rate. If the R to R is irregular. Count the number of R's and multiply by 10. Michael V Hyland: I'm not going to make you count the atro rate, so I'm not going to talk about it. Just know that the the more that gets blocked the lower the ventricular rate. the more life-threatening the rhythm becomes.

17 Rules for Type II Second Degree Heart Block (continued)

1:34:23 – 1:35:02

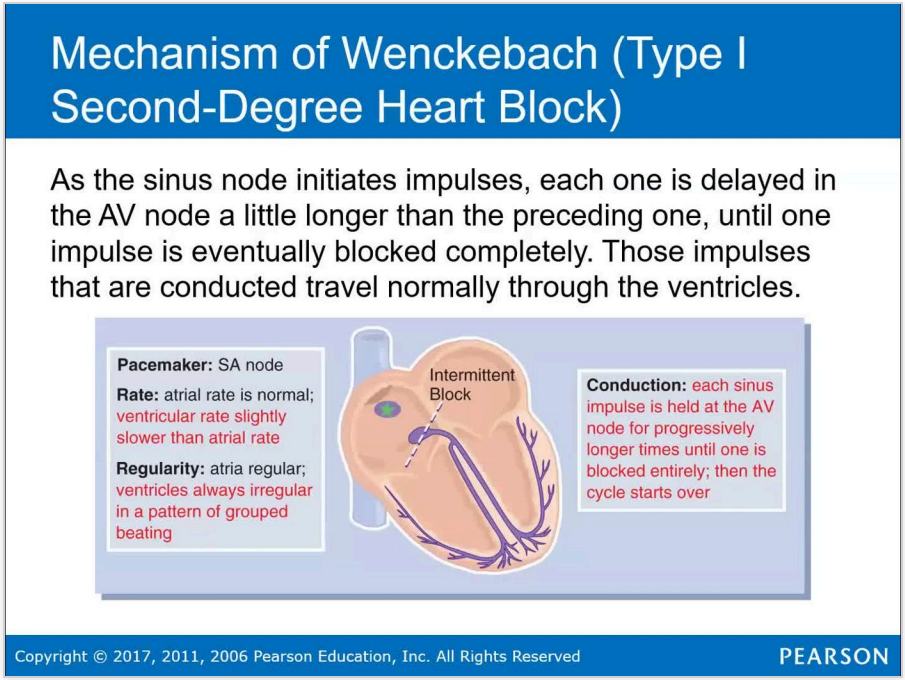


1:34:25 Michael V Hyland: P. Waves will be upright in uniform. There'll always be more P's than Q's, because they get blocked. The Pri on the conducted beats will be constant across the strip underline that part highlight. It put an asterisk there. That's one of the keys that the Pr. I will be constant across the strip on the conducted beats. It might be normal, or it might be long.

1:34:53 Michael V Hyland: You Rs will be narrow or normal. Michael V Hyland: and we already covered Winkebach.

18 Mechanism of Wenckebach (revisited)

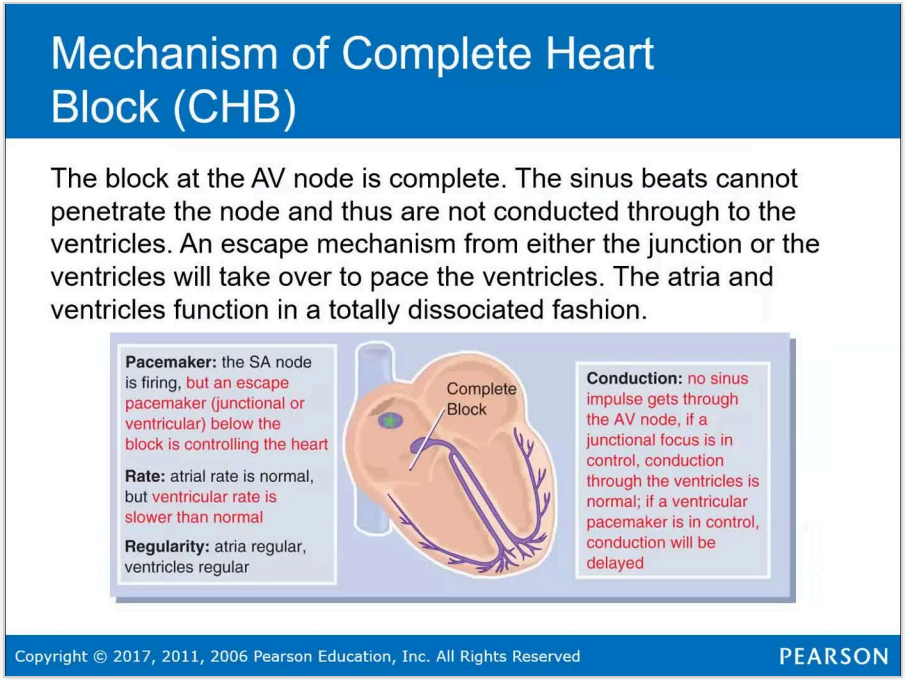
1:35:02 – 1:35:30



1:35:04 Michael V Hyland: and you don't have to interpret the underlying rhythm for second degree type one. It's just second degree type, one for second degree, type 2. It's just second degree type 2 for the interpretation. Michael V Hyland: It's first degree heart block that you have to uninterpret the underlying rhythm along with the block, which is not a true block. Just the delay. Michael V Hyland: all right, and the last one. thank God!

19 Mechanism of Complete Heart Block (CHB)

1:35:30 – 1:36:52



1:35:31 Michael V Hyland: So it talked out here. so mechanism for complete heart block. Michael V Hyland: So this is known as Third Degree heart block

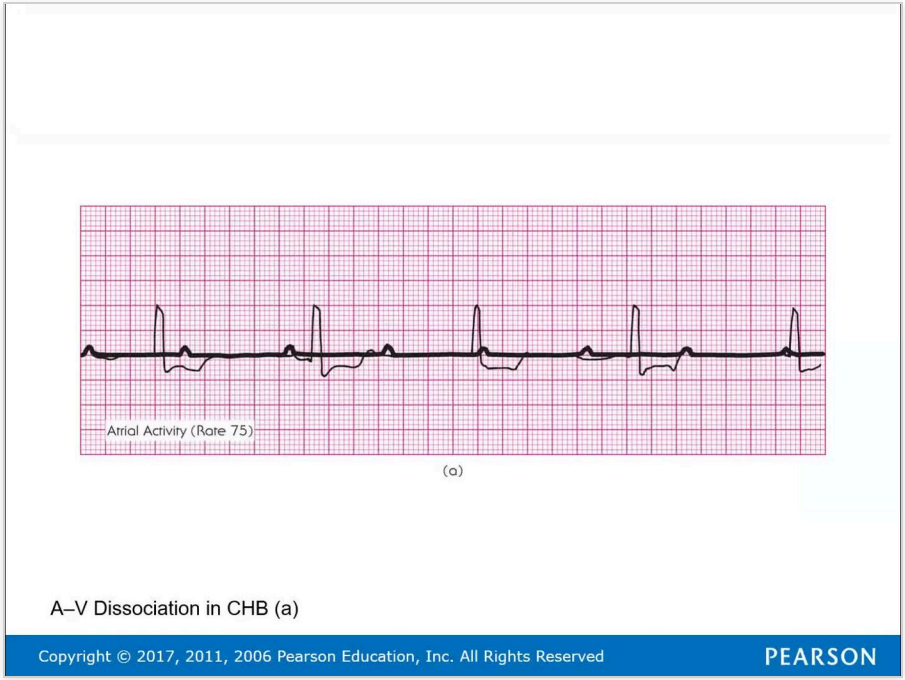
1:35:45 Michael V Hyland: or complete Heart Block. Michael V Hyland: or a the capital letter a Michael V Hyland: capital letter V. Dissociation. So Michael V Hyland: third degree, heart block, complete heart, block, or AV dissociation. Michael V Hyland: In this rhythm the A b node is completely blocked nothing. Michael V Hyland: no impulses penetrate the A B node.

1:36:16 Michael V Hyland: So, therefore, an escape mechanism from either the lower part of the junction Michael V Hyland: or somewhere in the ventricles needs to take over to pace the ventricles.

1:36:29 Michael V Hyland: the atria and ventricles function in a totally dissociated fashion. Michael V Hyland: so the atria are following Michael V Hyland: the essay notes, impulses. Michael V Hyland: the ventricles are following the AV nodes, impulses, or the ventricle impulses. So let's look at this.

20 A–V Dissociation in CHB (a, b)

1:36:52 – 1:38:15



1:36:55 Michael V Hyland: So the bold line is atrial activity. The thin line is ventricular activity. Michael V Hyland: So you can see that all of the bold line, all of these P. Waves there is no Qrs that follows it.

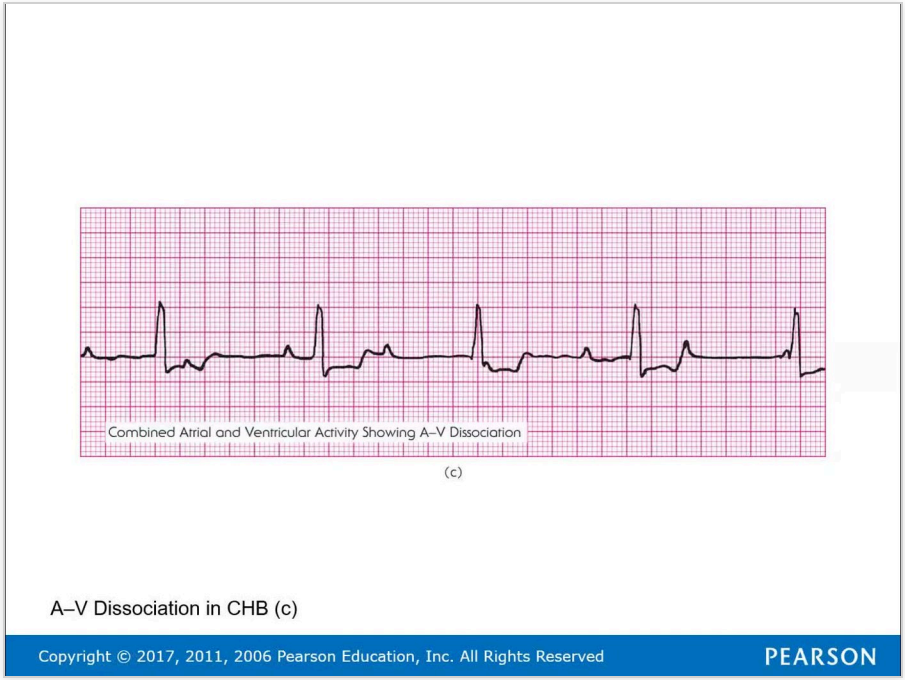
1:37:13 Michael V Hyland: It may eventually look like it's followed by a QRS. Way down here. But there's no relationship between this P. And this Q. Rs. and then I see another P wave. Michael V Hyland: and it's going to be difficult to see this P. Wave, because it's going to be inside of this beats T. Wave the ventricular beats in this rhythm have an inverted T-wave. They're upside down. Michael V Hyland: I see another P wave, followed by a QRS. But again, there's no relationship here. How do I know there's no relationship between P's and Q's is, look at the PRI. This one is super long.

1:37:52 Michael V Hyland: This one is in the T wave, and then there's another one before this. Qrs. This one is shorter, this one is longer. Michael V Hyland: this one is shorter, but still long. Right. and then, if I flip-flop the bold line. Now you can see the Q. Rss. As the bold and the p-waves as the thing.

1:38:14 Michael V Hyland: So what I really want to do is go to the entire rhythm here. So we need to collect data on this rhythm. So you guys know how to interpret this.

21 A–V Dissociation in CHB (c)

1:38:15 – 1:44:23



1:38:24 Michael V Hyland: So let's go through our information gathering Michael V Hyland: 510-15-2025, Michael V Hyland: 4 is 2920, 31 and a half. Michael V Hyland: 51015202530, 31, 32, 51015202530, Michael V Hyland: one and a half. 510-15-2025, 29, 30, 31, 32. So this rhythm is regular.

1:39:06 Michael V Hyland: 1,500, divided by 32. Michael V Hyland: Give me a heart rate in the 40 s. High, 40 s. 47. Michael V Hyland: So the rhythm is regular, the heart rate is 47. We need to keep in mind the heart rate because the heart rate's going to be one of the big clues. What pacemaker has a heart rate in the 40 s. Michael V Hyland: AV, no, right, it's 40 to 60. So we're gonna keep that in mind. Remember that AV note is completely blocked. We haven't discovered that yet. We're gonna discover that in a second. Michael V Hyland: Right? Michael V Hyland: So let's look at the P. Waves. We see a P. Wave. Here Michael V Hyland: we see another P wave in the middle of the T wave Michael V Hyland: P. Wave P. Wave. There's a P. Wave right here on the bottom of this. Qrs, there's another P wave here.

1:40:03 Michael V Hyland: There's a P. Wave right here on top of this T wave, and then another P wave right on the top of the beginning of this QRS. Michael V Hyland: So if you actually map that out, P to P would be regular.

1:40:18 Michael V Hyland: So the Sa note is firing, and the atria are firing at regular or contracting at regular intervals. Now we got to look at the Pri. So the P. Waves are upright in uniform. Michael V Hyland: Now we gotta look at the P. Ri. look at this 1, 5, 10

1:40:37 Michael V Hyland: and 4 and a half. That's 14 and a half. Let's look at the next one. Michael V Hyland: 5, Michael V Hyland: 6 and a half. That's 0 point 2 6. Let's look at the next 1, 5, 1014, Michael V Hyland: 1516, 1718. Let's look at the next one. Michael V Hyland: 5 and 4 make 9, Michael V Hyland: 10. Let's look at the next one. This pea waves right on top of it. So that's going to be 2 small boxes. Michael V Hyland: So there is such a wide Michael V Hyland: variety of number of small boxes between the P. Wave Michael V Hyland: that precedes each of these. QRS.

1:41:27 Michael V Hyland: What does that tell you Michael V Hyland: that tells you that there is no relationship between the P's and the Q's. So what do I expect you to write for the for the PRII expect you to write. Dissociated or not related. Michael V Hyland: The P's and Q's are not related. I know they're not related Michael V Hyland: because of how far off each of the pris are. There's no rhythm that has this kind of pri spread. Michael V Hyland: so it tells me the atria are doing their own thing, and the ventricles are doing their own thing.

1:42:05 Michael V Hyland: So right now, I know that this is a third degree block. Michael V Hyland: Everybody understand how I got there. Michael V Hyland: not hearing a lot of confidence.

1:42:19 Michael V Hyland: Okay, now, I'm going to measure the Q. Rs. so I'm going to measure from here back to the beginning. So 1, 2, maybe 2 and a half. Michael V Hyland: 2 and a half. So this QRS. Is point 1 0.

1:42:39 Michael V Hyland: Now, I told you, when we discover the patient has third degree heart block, which, by the way is life threatening immediately, life threatening? So when we discover the patient has third degree heart block, there's only going to be one of 2 pacemakers that pace the lower 2 chambers Michael V Hyland: which we already talked about in the rules. It's either going to be the junction or it's going to be the ventricles. Michael V Hyland: Do the ventricles have a wide or a narrow QRS. I'm sorry. Do that. Does the junction. If the junction, if this is a junctional escape rhythm? Is that a wide QRS. Or a narrow QRS.

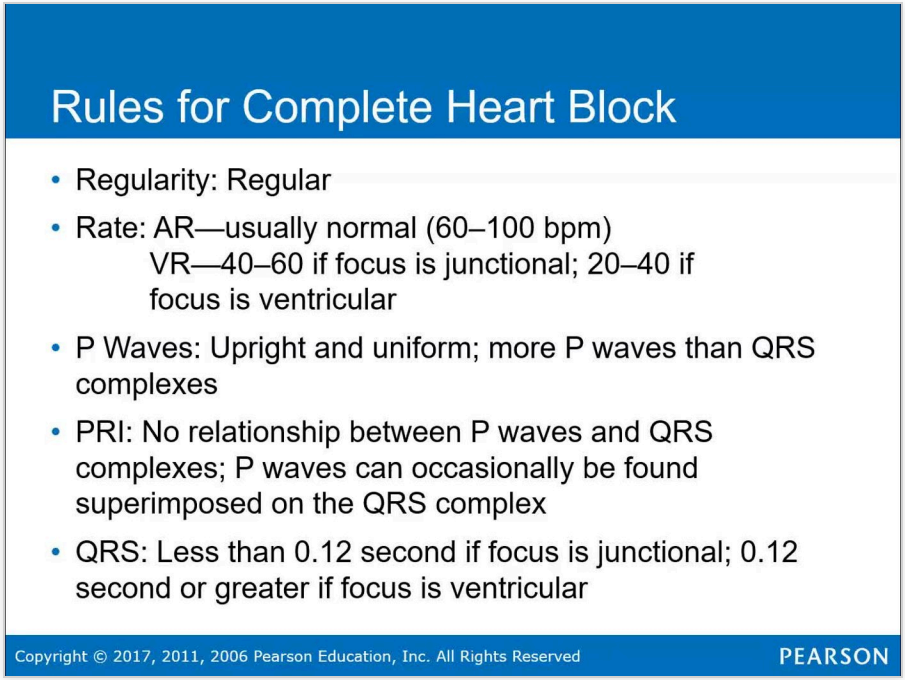
1:43:19 Michael V Hyland: Well do the ventricles get depolarized through the normal pathway? If it's the lower half of the junction. It's going to send it down the vulnerable branches to the Burkini fibers. So depolarization is gonna happen quickly and the Qrs will be narrow Michael V Hyland: and the inherent rate for the junction is 40 to 60. So we have a rate of 47 with a narrow QRS. So, gathering this information and no relationship between P's and Q's. This is third degree, heart block

1:43:51 Michael V Hyland: with a junction escape rhythm. Michael V Hyland: This is where it gets challenging. Now, if these Q. Rss. Were wide Michael V Hyland: point 1 2 s, or greater, and the heart rate was between 20 to 40. It would be third degree heart block with a ventricular escape, rip.

1:44:18 Michael V Hyland: So let's just quickly cover the rules, and then we'll end it for today.

22 Rules for Complete Heart Block

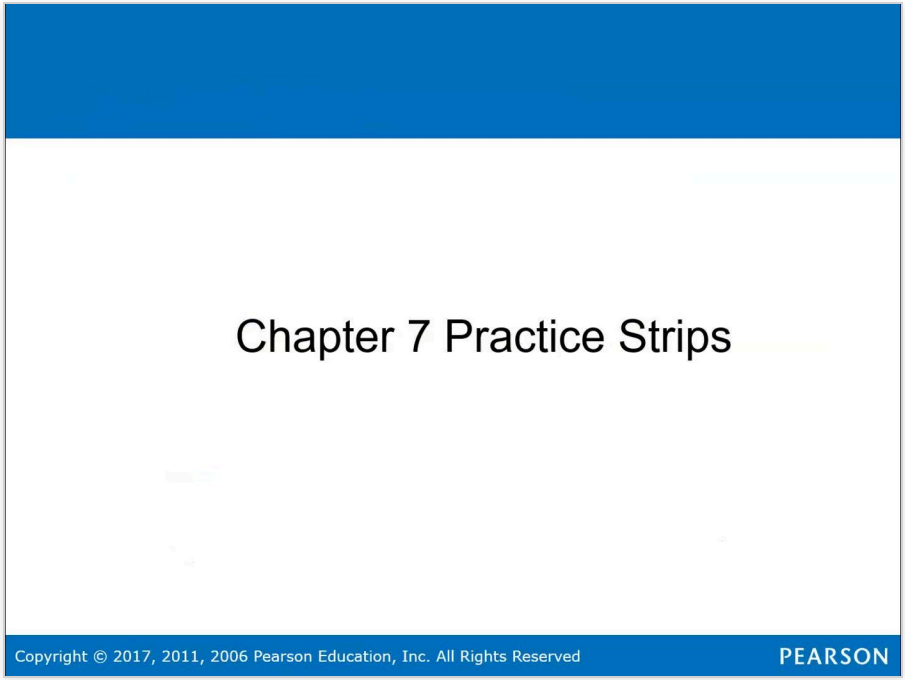
1:44:23 – 1:45:04



1:44:23 Michael V Hyland: So it's always regular. The atrial rate will always be normal. The ventricular rate will be 40 to 60 with a narrow Qrs. If the junction is pacing the lower 2 chambers. Michael V Hyland: the heart rate will be 20 to 40 with a wide Qrs. If it's a ventricular escape, rhythm. Michael V Hyland: P. Waves will be upright in uniform. The Pri. The P. Waves will not be related to the Q. Rss. Michael V Hyland: As I said, the QRS. Will be narrow if it's junctional. And why, if it's ventricular.

23 Chapter 7 Practice Strips

1:45:04 – 1:45:58



1:45:05 Michael V Hyland: hey, guys. Michael V Hyland: we're gonna review this again next week. Do the best you can with the homework. If you have questions, you can send me an email. I mean.

1:45:20 Michael V Hyland: no. I was wondering if I can email you my homework? Yes, you can. You can. Michael V Hyland: Alright. Thank you. What's your other question? Michael V Hyland: Alright!